Local Search for Subset Selection with Stochastic Feasibility Feedback

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Shared systems such as spectrum-sharing networks and compute clusters must often decide which subset of requests to admit together. Admitting more requests increases throughput, but each one adds an unknown amount of load to a shared budget. We consider a more stringent feedback setting in which the system reports only whether a configuration succeeded in a given epoch—not which requests contributed to a failure or by how much. We model this as follows: each of n items/jobs carries a hidden nonnegative risk; a subset S succeeds with probability $\theta(S) = f(R(S))$, where $R(S)$ is its total risk and f is an unknown non-increasing function; and each test of S returns a single Bernoulli$(\theta(S))$ outcome. The goal is to maximize the expected throughput $|S| \cdot \theta(S)$. This is a combinatorial bandit problem with non-linear bandit feedback; the action space is exponential and the objective has no simplifying structure such as submodularity.

We give local-search algorithms with approximation guarantees, under two assumptions on f. When f is only assumed non-increasing (so sharp jumps are allowed), a *swap-and-grow* procedure that exchanges up to t items per step achieves a $\frac{t}{t+1}$ approximation, up to an arbitrarily small additive error $\epsilon > 0$, using $n^{O(t)}$ samples; we show that $n^{\Omega(t)}$ queries are necessary for beating this factor even with exact access to θ. When f is strictly decreasing and its slope varies "smoothly", a *grow-and-tune* procedure finds a set within any additive ϵ of the optimal expected throughput using polynomially many samples. Both approaches extend to the online setting with $\tilde{O}(\sqrt{T})$ regret: $\frac{t}{t+1}$-approximate regret in the first case and standard regret in the second.

1 Introduction

Shared computing and networking systems are often required to solve an *admission-control* problem: which subset of requests can be admitted together? A cluster manager must decide which jobs to co-locate on a machine [11, 18, 19]; a spectrum coordinator must decide which transmitters may share a channel [41]. Admitting more requests increases potential throughput, but each admitted request also contributes load or interference to the shared resources or its system-wide budget, and exceeding that system limits can cause performance degradation or a protection violation. We consider a more stringent feedback setting in which the controller observes neither the contribution of each individual admitted item, nor even the total load. Instead, it receives only a binary indication of whether the configuration satisfied an aggregate-load threshold.

A motivating example is spectrum sharing in the vicinity of radio astronomy facilities. Radio telescopes are extremely sensitive receivers and have traditionally been protected by quiet zones in which transmissions are heavily restricted [40]. But new initiatives such as National Radio Dynamic Zones (NRDZ), a part of NSF's Spectrum Innovation Initiative [41], seek to replace such static exclusion with dynamic sharing: nearby secondary users are permitted to transmit as long as the aggregate interference at the protected receiver remains below a threshold. Suppose n transmitters request access, and transmitter i contributes an unknown nominal interference r_i at the receiver, so that admitting a set S produces nominal aggregate interference $R(S) = \sum_{i \in S} r_i$. In *our restricted-feedback setting*, the coordinator observes neither the individual contributions nor $R(S)$; after each epoch, it learns only whether the protection criterion was satisfied. This outcome is also stochastic: due to fading, background interference, and measurement noise, the same set may pass in one epoch and fail in the next [2]. Similar resource-sharing challenges appears in applications such as cluster co-location, network slicing [4, 19].

Motivated by these settings, we study the following abstraction. Each of n items has an unknown nonnegative risk r_i, the aggregate risk of a set S is $R(S) = \sum_{i \in S} r_i$, and a set S succeeds with probability $\theta(S) = f(R(S))$, where f is an unknown non-increasing function. Each trial of S yields a single Bernoulli$(\theta(S))$ outcome. We adopt an all-or-nothing notion of successful service: if the

trial succeeds, all $|S|$ admitted items are served; otherwise, none are counted as successfully served. The goal is thus to maximize the expected throughput $\mathrm{Obj}(S) := |S| \cdot \theta(S)$, which captures the fundamental tradeoff in shared systems: larger sets yield more "utility" when they succeed, but succeed less often. We study this learning problem: how a controller can learn a near-optimal configuration from repeated noisy trials, both offline and while earning throughput online.

This problem can be viewed as a combinatorial bandit with full-bandit feedback and a non-linear reward [1, 49]. In contrast to semi-bandit formulations [10], the learner receives no item-level feedback and therefore cannot estimate the risks r_i individually. The problem presents three distinct difficulties. The first is *informational*: since feedback passes through f, it may reveal very little about $R(S)$. For instance, if f is a threshold function, two sets that always succeed are indistinguishable, even if one is much closer to the threshold than the other. The second is *combinatorial*: there are 2^n candidate sets, and neither the objective nor the success probability satisfies standard structural properties, such as submodularity, on which subset-selection methods typically rely. The third is *statistical*: $\theta(S)$ must be estimated from noisy binary samples, and estimation errors are most consequential near the feasibility frontier, which is precisely where good solutions lie.

Our algorithms must therefore rely only on the additivity of risk and the monotonicity of f and we show that these suffice to obtain interesting guarantees. Although individual risks can never be estimated, we show that just this structure suffices to construct local-search certificates: informally, if the current set is substantially smaller than a good reference set, then a small local move can increase its cardinality without decreasing its success probability. We evaluate our algorithms under two standard metrics. The first is *query complexity*: the number of noisy subset evaluations needed to identify, with high probability, a set whose expected throughput is close to (or within a given factor of) the optimum. The second is *regret*: when the controller must "earn throughput" while it learns, and we measure the cumulative loss over T epochs relative to the best fixed set. Our algorithms use polynomially many queries and achieve $\tilde{O}(\sqrt{T})$ regret.

1.1 Our Results

We study safe subset selection under sample-based aggregate feedback in two regimes, depending on the assumptions we make on the function f (and thereby θ). In each regime, we consider two questions: first, how well can we optimize the objective using only polynomially many noisy subset evaluations; and second, what regret can we guarantee when the learning procedure is run online over a horizon of T rounds?

Monotonicity alone: For our first set of results, we make only the minimal assumption that f is non-increasing; specifically, $R(S_1) \le R(S_2) \implies \theta(S_1) \ge \theta(S_2)$. We do not assume any smoothness or regularity, and hence this setting includes functions with abrupt changes, such as threshold functions. In this regime, we show that a local-search procedure that we call *Swap-and-Grow* achieves a $t/(t+1)$ approximation, up to an arbitrarily small additive error $\epsilon > 0$, using $n^{O(t)}/\epsilon^2$ noisy samples; see Theorem 3.1. We further show that the n^t factor in the query complexity is unavoidable using a lower-bound proof: we give an instance for which any possibly randomized algorithm that even has *perfect oracle access* to $\theta(S)$ must make (roughly) $n^{\Omega(t)}$ queries in order to beat a $\frac{t}{t+1}$ approximation. See Theorem 3.5.

We then consider the online version of the problem. Since polynomial-query optimization in this regime is only guaranteed up to $\alpha := t/(t+1)$, we measure performance using the standard α-regret benchmark. More importantly, we establish a stronger guarantee for the output of Swap-and-Grow: its objective exceeds the αOPT benchmark by a positive margin. This yields a final α-regret of $\mathcal{R}_\alpha(T) = \widetilde{O}(\sqrt{T})$; see Theorem 3.10.

Strict monotonicity and Gradient-Smoothness. Our second set of results considers the case where f has more structure. We assume that f is strictly decreasing and satisfies the local Gradient-Smoothness condition in Def. 2.2. We design *Grow-and-Tune* algorithm that combines greedy growth with one-swap local-search steps, and uses Gradient-Smoothness to control the effect of estimation errors. We show that, using polynomially many samples, it obtains an arbitrarily small additive error from the true optimum; see Theorem 4.2. Thus, with this additional regularity on f, we can go beyond the $t/(t+1)$ approximation barrier of the first regime.

Finally, we study this regime in the online setting. A direct explore-then-commit application of Grow-and-Tune gives only a $T^{3/4}$-type regret bound. We therefore develop an online variant, *Optimistic Grow-and-Tune*, which uses adaptive sampling to focus exploration on candidate sets that may still contain the optimal objective, rather than spending samples uniformly across all of them. This design strategy helps to avoid wasting samples on sub-optimal candidatess and yields $\mathbb{E}[\mathcal{R}(T)] = \widetilde{O}(\sqrt{T})$ regret for fixed instance parameters, up to additive terms independent of the horizon T; see Theorem 4.11.

1.2 Related Work

Our model is closely related to combinatorial bandits [8, 9]. Unlike semi-bandit methods, which observe item-level outcomes [15, 33], we receive only a single subset-level Bernoulli outcome. Even compared with full-bandit methods based on linear aggregate feedback [46], our feedback is more restrictive: the success probability is $\theta(S) = f(R(S))$, where both the aggregate risk $R(S)$ and the function f are unknown. Related work on nonlinear combinatorial rewards and unknown aggregation functions provides further connections [1, 49]. However, we must learn not only which items to select but also how many, since larger sets offer greater potential throughput at the expense of success probability. Comparison-oracle optimization is also related [13], but exact comparisons are unavailable under our noisy feedback model. Our online results connect to approximation regret [29, 44] and local-search-based guarantees [6, 24]. These settings have stronger local properties and therefore local improvements can be certified from assumptions on the objective. On the contrary, in our setting a suboptimal set need not have an improving neighbor under $\mathrm{Obj}(S)$. Our guarantees instead uses only minimum assumptions and properties the additive-risk structure and, in the stronger regime, Gradient-Smoothness. Finally, safe and constrained bandits [26] and learning-based resource allocation in networking systems [4, 32] are related, but their feedback models and performance criteria differ. Detailed comparisons are provided in Appendix F.

2 Model and Preliminaries

We consider a repeated subset-selection problem in which a controller admits a set of items and observes only whether the admitted set succeeded. We first define the model and objective (2.1), then the feedback model (2.2) and performance criteria (2.3), and finally the two structural assumptions on the safety function under which our results hold (2.4).

2.1 Model and Objective

Let $\mathcal{T}$ be a ground set of $n = |\mathcal{T}|$ items. Each item $i \in \mathcal{T}$ has a fixed, unknown risk $r_i \geq 0$, and a subset $S \subseteq \mathcal{T}$ has aggregate risk $R(S) := \sum_{i \in S} r_i$. In the spectrum-sharing setting of Section 1, r_i is the interference that transmitter i induces at the protected receiver, and $R(S)$ is the aggregate interference. The probability that S is safe is $\theta(S) := f(R(S))$, where $f : [0, \infty) \to [0, 1]$ is an unknown non-increasing function, which we call the success probability function. Neither the risks $\{r_i\}$ nor f is known to the learner. If a trial succeeds, all admitted items are served; otherwise, none

are served. The objective is the expected number of items served,

$$\mathrm{Obj}(S) := |S|\,\theta(S),$$

and we write $S^\star \in \arg\max_S \mathrm{Obj}(S)$ and $\mathrm{OPT} := \mathrm{Obj}(S^\star)$. The objective trades off cardinality against safety probability. $\mathrm{Obj}(\emptyset) = 0$, while $\theta(\mathcal{T})$ may be arbitrarily small, so the optimum is in general attained at a non-trivial intermediate set which admits more items while maintaining safety.

Remark 2.1. Obj(.) function is in general neither monotone nor submodular. Consequently, guarantees for greedy or submodular maximization do not apply directly.

2.2 Feedback Model

The learner observes neither r_i, $R(S)$, nor $\theta(S)$. A query of S returns an independent outcome $Y(S) \sim \mathrm{Bernoulli}(\theta(S))$, where $Y(S) = 1$ indicates that S was succeeded. Given m independent queries of S, the learner forms the empirical estimate $\hat{\theta}_m(S) := \frac{1}{m}\sum_{j=1}^{m} Y_j(S)$.

Feedback is therefore aggregate, as in *full-bandit* models, and it is related only through the unknown nonlinear map f. In particular, the item-level quantities r_i are not identifiable from the observations. When f is a threshold function, two sets with $\theta = 1$ are indistinguishable regardless of their aggregate risk.

2.3 Performance Criteria

We evaluate algorithms in two settings.

Offline: The learner queries sets of its choice and then outputs a single set $\hat{S}$. For $\alpha \in (0, 1]$ and $\epsilon > 0$, the algorithm is an (α, ϵ)-approximation if $\mathrm{Obj}(\hat{S}) \geq \alpha(\mathrm{OPT} - \epsilon)$ with probability at least $1 - 1/\mathrm{poly}(n)$. Its cost is its query complexity, the total number of Bernoulli queries it makes.

Online: The learner plays a set S_q in each round $q = 1, \ldots, T$ and earns $|S_q|$ if the trial succeeds, and zero otherwise, so its expected reward in round q is $\mathrm{Obj}(S_q)$. We compare its cumulative reward with an α fraction (approximation factor) of the best fixed set's:

$$\mathcal{R}_\alpha(T) := \alpha T \cdot \mathrm{OPT} - \mathbb{E}\Big[\sum_{q=1}^{T} \mathrm{Obj}(S_q)\Big].$$

For $\alpha = 1$ this is the standard regret $\mathcal{R}(T)$. We use $\alpha < 1$ in Section 3.4, corresponding to the approximation guarantee of Theorem 3.1.

2.4 Structural Assumptions on f

We study two levels of structure on the safety probability function f.

Weakly monotone regime (Section 3): We assume only that f is non-increasing, i.e., $R(S_1) \leq R(S_2) \Rightarrow \theta(S_1) \geq \theta(S_2)$. This is the minimal assumption under which aggregate risk is meaningful. It admits discontinuous safety probability functions, including the threshold function $f(x) = \mathbb{1}[x \leq \tau]$, for which feedback is least informative. With a lower bound proof, we show in section 3.3 that this regime forces a tradeoff between approximation factor and query complexity.

Strictly monotone regime (Seciton 4): Our second regime imposes additional structure. We assume that f is differentiable and strictly decreasing, thus by definition, for all x, we have $f'(x) < 0$ (for $x \geq 0$, which is the range of interest). Moreover, we also assume that the *gradient* of f does not vary too sharply in a small enough interval. This motivates the following formal property.

Definition 2.2 (Gradient-Smoothness (GS) Property)**.** Let f be a differentiable, strictly decreasing function on $[0, \infty)$, and let $\Delta > 0$ and $\gamma \in (0, 1]$. We say that f satisfies the Gradient-Smoothness (GS) property with parameters (Δ, γ) if in any interval $I = [a, a + \Delta]$, we have

$$\frac{\min_{x\in[a,a+\Delta]} |f'(x)|}{\max_{x\in[a,a+\Delta]} |f'(x)|} \geq \gamma.$$

Thus, within any interval of length at most Δ, the magnitude of the derivative cannot change by more than a factor $1/\gamma$. Informally, the property says that in any interval of length at most Δ, if the derivative is large enough at *some point* in the interval, the derivative is large *throughout* the interval. The GS property turns out to be (roughly) equivalent to the assumption that the derivative is Log-Lipschitz. It can also be related to the condition that $|f''| \leq c|f'|$, which is a lower order analog of the well-studied self-concordance condition.

The following lemma shows a simple consequence of the GS property that will be used repeatedly in our analysis.

Lemma 2.3. *Suppose f is a monotone decreasing function that satisfies the GS property with parameters (Δ, γ) and let $I = [a, a + \Delta]$ be an interval. Then for any $x < y < z \in I$, we have:*

$$f(y) \geq f(z) + \frac{\gamma(z-y)}{(z-x)}(f(x) - f(z)).$$

In other words, the lemma says that if f drops significantly by moving from x to z, then moving "back" to y, we can "reclaim" some of the f value. This will turn out to be crucial for our analysis. The proof follows directly from the Gradient-Smoothness property and the mean value theorem; see Appendix B.1. Finally, we invoke GS only on intervals of length $\Delta^\star := 2\max_{i\in S^\star} r_i$, the scale of a single item near the optimum, rather than globally.

3 Weakly Monotone Safety Functions

We first study the weakly monotone regime, in which f is assumed only to be non-increasing. In particular, f can be discontinuous or abrupt threshold-like functions. Consequently, the learner cannot rely on smooth changes in $\theta(S)$ or any quantitative information to infer which nearby set is better. As a result, exact optimization is too expensive in general, and our goal is an approximation guarantee using only polynomially many subset evaluations.

Our key idea is a local-search procedure called *swap-and-grow*. Starting from the empty set, the algorithm grows the cardinality by one in each level. In the *swap-and-grow* algorithm, at cardinality level ℓ, it may replace up to t elements of its current set and add one more element than it removes, thereby moving to cardinality $\ell+1$. Let $\alpha := \frac{t}{t+1}$. We show that this neighborhood is large enough to certify a α approximation to the optimal objective, while the required query-complexity remaining polynomial for fixed t.

Note that the parameter t contains a natural approximation-complexity tradeoff: the approximation factor $\frac{t}{t+1}$ can be pushed arbitrarily close to 1 by increasing the swap parameter t, at the cost of an $O(n^{2t+2})$ factor in query complexity. In Section 3.3 we show that an n^t-type factor in the query complexity is *unavoidable* to beat a $\frac{t}{t+1}$-approximation without additional regularity on f.

3.1 Swap-and-Grow Local-Search

We now present Swap-and-Grow (Algorithm 1), which proceeds through cardinality levels. Starting from $S_0 = \emptyset$, the algorithm maintains a representative set S_ℓ of size ℓ and constructs candidates of size $\ell + 1$ at each level ℓ. Thus over n levels, the algorithm builds a sequence $S_0, S_1, \ldots, S_n$ where $|S_\ell| = \ell$. The key operation is an *up-to-t swap-and-grow neighborhood* move. When growing a set B, the algorithm may remove $s \leq t$ of its current elements and replace them with $s + 1$ new elements. Thus increasing the cardinality by one, while the swaps allow the algorithm to correct

Algorithm 1: Swap-and-Grow Local Search

Input: Ground set $\mathcal{T}$ with $n := |\mathcal{T}|$, swap parameter $t \in \mathbb{N}$, accuracy parameter $\delta > 0$
Output: A subset $\widehat{S} \subseteq \mathcal{T}$
Set $m \leftarrow \frac{8}{\delta^2}\left((2t+6)\ln n + \ln(2(t+1))\right)$.
$S_0 \leftarrow \emptyset$;
for $\ell = 0, \ldots, n-1$ **do**
  $C_\ell \leftarrow \Gamma_t(S_\ell)$;
  foreach $S \in C_\ell$ **do**
    Estimate $\widehat{\theta}(S)$ using m independent samples;
  $S_{\ell+1} \leftarrow \arg\max_{S \in C_\ell} \widehat{\theta}(S)$;
$\widehat{S} \leftarrow \arg\max_{\ell \in \{1,\ldots,n\}} |S_\ell| \widehat{\theta}(S_\ell)$;
return $\widehat{S}$;

earlier choices as it grows. Formally, define

$$\Gamma_t(B) := \{(B \setminus U) \cup V : U \subseteq B,\ V \subseteq \mathcal{T} \setminus B,\ |U| = s,\ |V| = s+1,\ 0 \le s \le \min\{t, |B|\}\}.$$

Thus every set in $\Gamma_t(B)$ has cardinality $|B|+1$. Notice that $s = 0$ corresponds to a pure grow step, whereas $s > 0$ allows the algorithm to revise up to t previously selected items while advancing to the next cardinality level. At level ℓ, the algorithm evaluates every candidate in $C_\ell := \Gamma_t(S_\ell)$ using m Bernoulli samples and chooses the candidate with the largest empirical safety probability $S_{\ell+1} \in \arg\max_{S \in C_\ell} \widehat{\theta}(S)$. After completing all levels, the algorithm returns the subset among $\{S_1, \ldots, S_n\}$ with the largest empirical objective $|S_\ell|\widehat{\theta}(S_\ell)$.

Complexity of Alogrithm 1: For fixed t, the neighborhood remains polynomial in n. Indeed, $|\Gamma_t(S_\ell)| = \sum_{s=0}^{\min\{t,\ell\}} \binom{\ell}{s}\binom{n-\ell}{s+1}$, and hence the total number of candidate evaluations over all levels is at most $M_t := (t+1)n^{2t+2}$. We use M_t below when establishing uniform concentration over all evaluations made by the algorithm.

3.2 Approximation Guarantee

We now establish the approximation guarantee of Algorithm 1. Our analysis has two main ingredients: Firstly, we show that, in the cardinality level ℓ, if the current representative S_ℓ is sufficiently smaller than a reference set A, then its swap-and-grow neighborhood C_ℓ contains a candidate whose safety probability is no worse than the smaller of $\theta(S_\ell)$ and $\theta(A)$. Secondly, a uniform concentration event ensures that empirical estimates are accurate enough. Chaining these two observations across cardinality levels shows that the algorithm approximately preserves the safety of the good reference set A while growing to roughly a $t/(t+1)$-fraction of its cardinality. Throughout this section, let $\alpha := \frac{t}{t+1}$, and recall that a complete run of Algorithm 1 evaluates at most $M_t := (t+1)n^{2t+2}$ candidate subsets. Our main result here is the following.

Theorem 3.1. *Let $S^\star \subseteq \mathcal{T}$ maximizes $Obj(S)$, and let $\epsilon > 0$. Run the algorithm 1 with $\delta := \frac{\alpha\epsilon}{2n^2}$. Then with probability at least $1 - \frac{1}{n^4}$, its output $\widehat{S}$ satisfies:*

$$Obj(\widehat{S}) \ge \frac{t}{t+1} \cdot (Obj(S^\star) - \epsilon).$$

Moreover, the number of noisy queries made by the algorithm is $poly(n^t)/\epsilon^2$.

We begin with the following structural lemma that explains why the swap parameter t leads to the factor $t/(t+1)$.

Lemma 3.2. *Assume $r_i \geq 0$ for all $i \in \mathcal{T}$. Let $A, B \subseteq \mathcal{T}$, with $A \neq \emptyset$, and suppose $|A| \geq \frac{t+1}{t}|B|$. Then there exists a set $B^+ \in \Gamma_t(B)$ such that*

$$R(B^+) \leq \max\{R(A), R(B)\}.$$

Consequently, since f is non-increasing, $\theta(B^+) \geq \min\{\theta(A), \theta(B)\}$.

Proof Overview of Lemma 3.2: The idea is to replace high-risk elements of B with low-risk elements from A, while increasing the cardinality by one. If $B \subseteq A$, a simple grow step suffices. Otherwise, let $d := |B \setminus A|$ and $s := \min\{t, d\}$. The cardinality assumption ensures that $A \setminus B$ contains at least $s+1$ elements. We remove the s largest-risk elements of $B \setminus A$ and add the $s+1$ smallest-risk elements of $A \setminus B$, obtaining $B^+ \in \Gamma_t(B)$.

By averaging, the removed risk is at least $\lambda R(B \setminus A)$, while the added risk is at most $\lambda R(A \setminus B)$, where $\lambda := s/d \in (0, 1]$. Thus, $R(B^+) \leq (1-\lambda)R(B) + \lambda R(A) \leq \max\{R(A), R(B)\}$.

Since f is non-increasing, the claimed bound on $\theta(B^+)$ follows. The complete proof is given in Appendix C.1. Our Algorithm 1 chooses candidates using empirical estimates, therefore, next, we show that for our choice of m (samples per set), we obtain sufficiently accurate estimates at all times.

Lemma 3.3. *Let $\delta > 0$ be given. Consider Algorithm 1, which evaluates a sequence of candidate sets $S_1, S_2, \ldots, S_N$ chosen adaptively during its execution. (Formally, at any q, the choice of S_q depends on the sets S_i and the estimates $\widehat{\theta}(S_i)$, for $i < q$.) Then if each query is evaluated using $m = \frac{8}{\delta^2}((2t+6)\ln n + \ln(2(t+1)))$ samples, with probability at least $1 - \frac{1}{n^4}$, we have*

$$|\theta(S_q) - \widehat{\theta}(S_q)| < \frac{\delta}{4} \quad \text{for all } 1 \leq q \leq N.$$

Let us denote this event by $\mathcal{E}$. (The proof follows from Hoeffding's inequality together with total subsets queried $M_t \leq (t+1)n^{2t+2}$, and is deferred to Appendix A.2.)

The next lemma is crucial to the analysis. It shows that a high *safety probability* "persists" across cardinality levels of the swap-and-grow algorithm. In what follows, let us write $\alpha = \frac{t}{t+1}$.

Lemma 3.4. *Assume that $\mathcal{E}$ from Lemma 3.3 holds. Fix a non-empty reference set $A \subseteq \mathcal{T}$. For every cardinality level ℓ satisfying $|A| \geq \frac{t+1}{t}\ell$, the representative chosen by algorithm 1 satisfies*

$$\theta(S_{\ell+1}) \geq \min\{\theta(A), \theta(S_\ell)\} - \frac{\delta}{2}. \tag{1}$$

Furthermore, suppose we define $L_A := \lfloor \alpha|A| \rfloor + 1$. Then representatives $S_0, S_1, \ldots, S_n$ constructed by Algorithm 1 satisfies $\theta(S_\ell) \geq \theta(A) - \frac{\delta\ell}{2}$ for every $0 \leq \ell \leq L_A$.

Proof Overview of Lemma 3.4: For any level ℓ satisfying the cardinality condition, Lemma 3.2 guarantees that there exists a candidate in $\Gamma_t(S_\ell)$ whose safety probability is at least $\min\{\theta(A), \theta(S_\ell)\}$. Since Algorithm 1 selects the candidate with the largest empirical safety probability, the uniform concentration guarantee of Lemma 3.3 implies that we lose at most an additional $\delta/2$, establishing (1). The persistence bound then follows by induction. Initially, $\theta(S_0) \geq \theta(A)$ since $S_0 = \emptyset$. Moreover, the cardinality condition continues to hold for every $\ell < L_A$. Thus, repeatedly applying (1) gives $\theta(S_\ell) \geq \theta(A) - \frac{\delta\ell}{2}$, for every $0 \leq \ell \leq L_A$. The complete proof is given in Appendix C.1.

We now joint the arguments to prove Theorem 3.1:

Proof Overview of Theorem 3.1. Let $S^\star$ be an optimal set and set $k := \lceil \alpha|S^\star| \rceil$. Conditioning on the uniform concentration event of Lemma 3.3, we apply Lemma 3.4 with $A = S^\star$ to obtain $\theta(S_k) \geq \theta(S^\star) - \frac{k\delta}{2}$.

Since $k \geq \alpha|S^\star|$, this gives $\mathrm{Obj}(S_k) \geq \alpha\mathrm{OPT} - \frac{\delta n^2}{2}$. Finally, Algorithm 1 returns the set with the largest empirical objective among all the constructed representatives. By Lemma 3.3, this final selection loses at most an additional $n\delta/2$. With the prescribed choice of δ, the total error is at most $\alpha\epsilon$, giving the claimed guarantee. The complete proof is given in Appendix C.1.

□

3.3 Lower Bound

It is natural to ask if the n^t dependence is necessary to beat a $t/(t+1)$ approximation to the objective. Without any additional assumptions on f, we show that even in the simplest setting of θ being defined by a threshold, beating this approximation requires $\Omega\left(\binom{n}{t+1}\right)$ queries in the worst case.

Theorem 3.5. *For any fixed $t \geq 1$ and all sufficiently large n, any randomized algorithm $\mathcal{A}$ with query complexity $< \frac{1}{10}\binom{n}{t+1}$, there exists a weakly monotone set function θ such that: (i) for the optimal set $S^\star$, we have $Obj(S^\star) \geq (t+1)$, and (ii) with probability at least $3/4$, the output S of the algorithm satisfies $Obj(S) \leq t$.*

Proof Overview of Theorem 3.5. We use Yao's minimax principle. The idea is to hide a uniformly random set J of size $t+1$, assign risk t to its elements and risk $t+1$ to all remaining elements, and use the threshold $t(t+1)$. Under this construction, every set of size at most t succeeds, while every set of size at least $t+2$ fails. Among sets of size $t+1$, only J succeeds. Thus, unless an algorithm queries J, the feedback reveals no information about its identity.

Consequently, a deterministic algorithm making q queries can identify J with probability at most $(q+1)/\binom{n}{t+1}$, which is at most $1/4$ under the stated query budget for sufficiently large n. The claimed lower bound then follows from Yao's principle. The complete proof is given in Appendix C.2. □

3.4 Online Learning: α-Regret

We next consider the online version of the problem. Over a horizon of T rounds, the learner selects a subset $S_q \subseteq \mathcal{T}$ at each round q, receives a Bernoulli success observation, and earns expected reward $\mathrm{Obj}(S)$. Let $\mathrm{OPT} := \max_{S\subseteq\mathcal{T}} \mathrm{Obj}(S)$ and $\alpha = \frac{t}{t+1}$. Since Swap-and-Grow guarantees an α-approximation under weak monotonicity, we measure performance against the corresponding approximation benchmark. Define the expected α-regret as, $\mathcal{R}_\alpha(T) := \alpha T \cdot \mathrm{OPT} - \mathbb{E}[\sum_{q=1}^{T} \mathrm{Obj}(S_q)]$. Thus, each round is compared against an α-fraction of the optimal expected reward. In particular, a set whose objective exceeds $\alpha\mathrm{OPT}$ incurs negative α-regret; this slack will be useful below.

A direct explore-then-commit application of Theorem 3.1 already gives a sub-linear α-regret. This is because Algorithm 1 achieves an objective of at least $\alpha(\mathrm{OPT} - \epsilon)$ using $O(1/\epsilon^2)$ noisy samples per candidate, Thus the regret can be bounded by (ignoring n factors) $O(\frac{1}{\epsilon^2}) + \epsilon T$, and by setting ϵ appropriately, we can achieve an α-regret of $T^{2/3}$.

In this Section, we show how to improve upon this simple procedure, obtaining an α regret of $\widetilde{O}(\sqrt{T})$ by exploiting a small slack hidden in the proof of Theorem 3.1. If we carefully observe Lemma 3.4 with the similar notations used, for a large reference set A, it is guaranteed that the algorithm chooses sets S_ℓ of size $\lfloor\alpha|A|\rfloor$ as well as set $S_{\ell+1}$ of size $\lfloor\alpha|A|\rfloor + 1$ who have success probabilities $\theta(.)$ approximately as the same as $\theta(A)$. Now, this gives us a leverage. Consider an example, where the reference set A is the the optimal set. We also consider $t = 2$, $|A| = 6$ and sets S_ℓ and $S_{\ell+1}$ to be S_4 and S_5 respectively. Here, we have $\alpha|A| = 4$ while guarantee continues through cardinality level 5. Thus ignoring estimation error, the algorithm obtains objective about $5\theta(A)$ with the set S_5, whereas the factor-α benchmark was only $4\theta(A)$. With some careful sampling, this additional margin (objective surplus) gives us α-regret ≤ 0 and thus can absorb the error incurred in the early exploration. We formalize this argument in the following lemma:

Lemma 3.6. *Let $\alpha := \frac{t}{t+1}$. Fix a nonempty reference set $A \subseteq \mathcal{T}$, and define $L_A := \lfloor \alpha|A| \rfloor + 1$. Suppose the event $\mathcal{E}$ from Lemma 3.3 holds. Then*

$$Obj(S_{L_A}) \geq \alpha Obj(A) + \frac{\theta(A)}{t+1} - \frac{\delta}{2}n^2.$$

Consequently, the final output of Algorithm 1 satisfies $Obj(\widehat{S}) \geq \alpha Obj(A) + \frac{\theta(A)}{t+1} - \delta n^2$.

Proof of Lemma 3.6. Recall that Lemma 3.4 implies that $\theta(S_{L_A}) \geq \theta(A) - \frac{\delta}{2}L_A$. Using this, we get $\mathrm{Obj}(S_{L_A}) = L_A \cdot \theta(S_{L_A}) \geq L_A\theta(A) - \frac{\delta}{2}L_A^2 \geq L_A\theta(A) - \frac{\delta}{2}n^2$.

Now the key observation is that $L_A - \alpha|A|$ is always $\geq \frac{1}{t+1}$. This is because by definition, if $\alpha|A|$ is an integer, then $L_A = \alpha|A| + 1$, and so the difference is actually 1. If $\alpha|A|$ is a fraction, i.e., $\alpha|A| = \lfloor \alpha|A| \rfloor + \zeta$ for some $\zeta \in (0,1)$, then because ζ is an integer multiple of $\frac{1}{t+1}$, we must have $\zeta \leq 1 - \frac{1}{t+1}$. Thus $L_A - \alpha|A| \geq \frac{1}{t+1}$. This implies that $L_A\theta(A) \geq \alpha\mathrm{Obj}(A) + \frac{\theta(A)}{t+1}$, which proves the first claim. For the final output, the same empirical-selection argument used in the proof of Theorem 3.1 loses at most $n\delta/2$. Since $n\delta/2 \leq \delta n^2/2$, $\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{Obj}(A) + \frac{\theta(A)}{t+1} - \delta n^2$. □

This is the central insight. Now, if we consider $A = S^\star$, under $\mathcal{E}$, the committed set $\widehat{S}$ by algorithm 1 satisfies $\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{OPT}$ as long as $\delta < \mathrm{OPT}/n^3$. The remaining challenge is how to set the accuracy parameter δ for algorithm 1 as we don't know $\theta(S^\star)$ or OPT.

3.4.1 Estimating Optimum via Singletons. In this section, we try to address the challenge addressed above by introducing an observable scale that we call the *singleton-scale* of the instance. Formally, $g := \max_{i \in \mathcal{T}} \theta(\{i\})$.

The following lemma is almost immediate:

Lemma 3.7. *Let $S^\star$ be an optimal set, and write $\theta^\star := \theta(S^\star)$. If $r_i \geq 0$ for all $i \in \mathcal{T}$, then*

$$g \leq \mathrm{OPT} = |S^\star|\theta^\star \leq ng, \qquad \text{and consequently} \qquad \theta^\star \geq \frac{g}{n}.$$

The proof is given in Appendix C.3.

Thus g lower-bounds the surplus in Lemma 3.6 up to a factor n. Now, the issue is, we don't have access to the true value g. Therefore, next we show how we can build a simple estimator for g. The guarantee we obtain is captured by the following Lemma.

Lemma 3.8. *Let $g_0 \in (0,1)$ be a given parameter, and let $g := \max_{i \in \mathcal{T}} \theta(\{i\})$ be the singleton-scale (which is unknown) of the given instance. There exists an algorithm that makes noisy queries as before, and outputs either a real number $\widehat{g}$ or* `small`*, such that the following hold with probability at least $1 - T^{-2}$:*

1. *If the algorithm returns* `small`*, then $g \leq 2g_0$.*
2. *If the algorithm returns a value $\widehat{g}$, then $\frac{g}{2} \leq \widehat{g} \leq 2g$ and also, $g \geq c_0 g_0$, for some absolute constant c_0.*

Finally, the number of queries made by the algorithm is at most $Q_{\mathrm{sing}} = \widetilde{O}\left(\frac{n}{\max\{g, g_0\}^2}\right)$.

Lemma 3.8 provides two possible outcomes: either the estimator certifies that the singleton scale is small, in which case $g \leq 2g_0$, or it returns an estimate $\widehat{g}$ within a factor of two of g. The procedure tests geometrically decreasing probability scales using singleton observations. We defer its implementation (Algorithm 6) and concentration analysis to Appendix C.3.

Algorithm 2: Singleton-Scale Explore-Then-Commit

Input: ground set $\mathcal{T}$, horizon T, swap parameter t.
Set $B_t := C(t+1)^3 n^{2t+9} \log(nT)$, for a sufficiently large universal constant C
Set $g_0 := \sqrt{\frac{B_t}{nT}}$.
if $g_0 \geq 1$ **then**
  Play $\emptyset$ for all T rounds and stop
Run SingletonScale(g_0, T) (Lemma 3.8)
if *SingletonScale returns* small **then**
  Play $\emptyset$ for all remaining rounds
else
  Let $\widehat{g}$ be the returned singleton-scale estimate
  Set $\xi := \frac{\widehat{g}}{64n^3(t+1)}$.
  Run Algorithm 1 with per-candidate sample count $m_\xi = \left\lceil \frac{\log(4M_t T^2)}{2\xi^2} \right\rceil$.
  Let $\widehat{S}$ be the set returned by Algorithm 1
  Play $\widehat{S}$ for all remaining rounds

3.4.2 Calibrated Swap-and-Grow. We now use the singleton-scale to run Algorithm 1 with a target comparison accuracy ξ. ξ is analogous to our earlier parameter δ. However, this time we are considering ξ as a function of singleton-scale g. Let $M_t := (t+1)n^{2t+2}$ be the total number of candidate evaluations in one complete run of Algorithm 1 parameter t. Now, we show the confidence event of Algorithm 1 with the target accuracy ξ and how it generates the "surplus" promised in the introduction of this section.

Lemma 3.9. *Fix $\xi > 0$. Suppose Algorithm 1 samples every candidate set $m_\xi := \left\lceil \frac{\log(4M_t T^2)}{2\xi^2} \right\rceil$ times. Then, with probability at least $1 - T^{-2}/2$, all empirical estimates used by Algorithm 1 are accurate to within ξ. Moreover, the total number of samples used by Algorithm 1 is at most*

$$Q_{\mathrm{SG}}(\xi) \leq M_t m_\xi = O\left(\frac{(t+1)n^{2t+2} \log(nT)}{\xi^2} \right).$$

Under the same event, if $\xi \leq \frac{g}{32n^3(t+1)}$, then the output $\widehat{S}$ of Algorithm 1 satisfies $\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{OPT} + \frac{g}{2n(t+1)}$. (The proof is given in Appendix C.4.)

3.4.3 Balancing Exploration and Commit. We now choose the baseline g_0 by balancing two regime. Here we give the intuition intuition first. Let B_t denote the polynomial and logarithmic factor in the exploration-cost bound, as specified in Algorithm 2. Now, recall SingletonScale estimator in Lemma 3.8. If the estimator returns small, then $g \leq 2g_0$ and so $\mathrm{OPT} \leq 2ng_0$, giving at most $O(ng_0 T)$ regret. Otherwise, the singleton estimator returns $\widehat{g} = \Omega(g_0)$, and Lemma 3.9 ensures that we regret ≤ 0 once we commit, therefore, in this case we only have exploration regret at most $\widetilde{O}(B_t/g_0)$. Balancing these two terms gives $g_0 := \sqrt{\frac{B_t}{nT}}$. This motivates our following "revised" explore-then-commit algorithm.

If the planned exploration exceeds the remaining horizon, the algorithm simply continues the exploration procedure until the horizon ends. The regret analysis below upper bounds the regret by the planned exploration cost, so this truncation can only improve the bound.

Theorem 3.10. *For $\alpha = \frac{t}{t+1}$, the α-regret over horizon T of Algorithm 2 satisfies*

$$\mathcal{R}_\alpha(T) = \widetilde{O}((t+1)^{3/2} n^{t+5} \sqrt{T})$$

Equivalently, for fixed t, $\mathcal{R}_\alpha(T) = \widetilde{O}_t(n^{t+5}\sqrt{T})$.

Proof Overview of Theorem 3.10. We consider the two possible outcomes of the singleton-scale estimator. If it returns `small`, Lemma 3.8 gives $g \le 2g_0$, and hence, by Lemma 3.7, $\mathrm{OPT} \le 2ng_0$. Thus, the regret over the entire horizon is at most $2ng_0T$.

Otherwise, Lemma 3.8 guarantees that the returned estimate $\widehat{g}$ is within a constant factor of g, with $g = \Omega(g_0)$. We can therefore calibrate Swap-and-Grow using $\widehat{g}$, and Lemma 3.9 guarantees that its output has objective at least $\alpha\mathrm{OPT}$. Consequently, the committed rounds incur nonpositive α-regret, and therefore we only need to bound the regret during exploration. Using the sample bounds of Lemmas 3.8 and 3.9, together with $\mathrm{OPT} \le ng$ from Lemma 3.7, this contribution is $\widetilde{O}(B_t/g_0)$. Finally, choosing $g_0 = \sqrt{B_t/(nT)}$ balances the two bounds and gives $\widetilde{O}(\sqrt{nB_tT})$, as desired. The remaining details are deferred to Appendix C.4. □

4 Strictly Monotone Safety Functions

We now turn to the strictly monotone regime. As before, the success probability of a set S is $\theta(S) = f(R(S))$, but now f is strictly decreasing. However, we have another stronger structural assumption of Gradient-Smoothness property with parameters (Δ, γ) as defined in Definition 2.2. Intuitively, GS property prevents the slope of f from changing too sharply over the local risk interval Δ. This gives the problem a stronger structure and therefore, we can go beyond the $t/(t+1)$ approximation guarantee. The precise value of Δ used in the analysis will be stated below.

We first explain an interesting structural consequence of strict monotonicity when we have access to the true-value of $\theta(S)$ through an *exact oracle* in Section 4.1. This is only a proof device which helps us later in the analysis of our original noisy-estimation setting using Bernoulli feedback. We then return to our sample-based feedback model and present its respective *Grow-and-Tune* algorithm in Section 4.2. We then analyze the algorithm and show how even under noisy feedback, the algorithm ensures an ϵ-close guarantee to the global optimum in Section 4.3. We also consider the performance of our algorithm in the online setting, and to achieve a sublinear regret of $\widetilde{O}(\sqrt{T})$ on the time horizon T, we present a modified algorithm in Section 4.4.

4.1 Structural Warm-Up with Exact Oracle Access

We first show how the stronger assumptions in the strictly monotone setting yield a useful property. For this subsection only, suppose the learner has *exact-oracle access*, in which, for any query of a subset $S \subseteq \mathcal{T}$, the oracle returns the true value $\theta(S)$. Our original feedback model is still the noisy Bernoulli feedback, whereas, we use this oracle-access setting only to show the property. At first, we make a simple observation. Suppose that the elements of $\mathcal{T}$ are ordered non-decreasing of risks such that $r_{j_1} \le r_{j_2} \le \cdots \le r_{j_n}$. For each cardinality level ℓ, we choose an optimal set as $S^\star_\ell = \arg\max_{S \subseteq \mathcal{T}, |S|=\ell} \theta(S)$, it is clear that $S^\star_\ell = \{j_1, j_2, \ldots, j_\ell\}$. In other words, the optimal sets at each "level" form a nested sequence, and therefore, a simple greedy growth procedure such as Algorithm 3 can be used to obtain the solution.

Algorithm 3: Greedy Growth with Exact Oracle Access

$S_0 \leftarrow \emptyset$
for $\ell \leftarrow 1$ **to** n **do**
  $i_\ell \leftarrow \arg\max_{i \in \mathcal{T} \setminus S_{\ell-1}} \theta(S_{\ell-1} \cup \{i\})$
  $S_\ell \leftarrow S_{\ell-1} \cup \{i_\ell\}$
$\widehat{\ell} \leftarrow \arg\max_{0 \le \ell \le n} \ \ell\, \theta(S_\ell)$
return $S_{\widehat{\ell}}$

Lemma 4.1. *For every $\ell \in \{0, ..., n\}$ and $S_\ell^\star$ defined as above, the set S_ℓ produced by the algorithm 3 satisfies $R(S_\ell) = R(S_\ell^\star)$. Consequently,*

$$S_0 \subset S_1 \subset \cdots \subset S_n.$$

is a nested chain of cardinality-level optimal sets. The proof of Lemma 4.1 is given in Appendix D.1.

Though under exact access to θ this nested structure makes the problem easy, under noisy Bernoulli's feedback small differences between candidate success probabilities cannot be resolved exactly. The next subsection approximately recovers the structure from noisy observations.

4.2 Grow-and-Tune Local Search

We now return to the noisy Bernoulli's feedback setting and introduce the *Grow-and-Tune* algorithm (Algorithm 4) that progresses through cardinality levels, maintaining a representative S_ℓ of size ℓ at each level. From a current representative set S_ℓ, at first it grows by adding one new element. Since the feedback is noisy, a single greedy addition may not be enough, therefore, after growing the set, the algorithm also performs local swaps to correct possible mistakes.

Suppose the current set is $S \subseteq \mathcal{T}$ with $|S| = \ell$. We use two neighborhood as follows:

Grow (cardinality increase). The grow neighborhood consists of all single-element additions: $\mathcal{G}(S) \coloneqq \{ S \cup \{j\} : j \in \mathcal{T} \setminus S \}$, where $|\mathcal{G}(S)| = n - \ell$. A grow move advances the algorithm from cardinality level ℓ to level $\ell + 1$.

Tune (within-level swap search). The tune neighborhood is the single-swap neighborhood, consisting of all replacements that preserve cardinality: $\mathcal{N}(S) \coloneqq \{ (S \setminus \{i\}) \cup \{j\} : i \in S,\ j \in \mathcal{T} \setminus S \}$,

where, $|\mathcal{N}(S)| = \ell(n - \ell)$. A tune move keeps the algorithm at cardinality level ℓ and is used to improve the current size-ℓ solution.

Algorithm 4 combines these two operations at each cardinality level. Starting from the empty set, it selects the empirically best Grow candidate and then repeatedly performs the best one-swap Tune move, provided its empirical improvement exceeds η. When no such improvement remains, the algorithm advances to the next cardinality level. Finally, it returns the representative with the largest empirical objective.

Since at most $1/\eta$ tune moves can be accepted at each level ℓ, Algorithm 4 uses $O(\frac{n^5}{\eta\delta^2\gamma^2} \log(\frac{n}{\eta})) = O(\frac{n^6}{\eta\delta^3\gamma^3} \log(\frac{n^2}{\delta\gamma}))$ Bernoulli samples in total, where the second bound uses $\eta = \delta\gamma/(8n)$.

4.3 Approximation Guarantee

We now analyze Algorithm 4 under the strictly monotone setting. The proof combines three ingredients: (i) a uniform concentration event ensuring that all empirical comparisons used by the algorithm are reliable, (ii) a structural lemma that to show how *swap*-move reduces accumulated risk ($R(.)$), and (iii) the Gradient-Smoothness property (via Lemma 2.3) to convert improvements in risk ($R(.)$ scale) into improvements in safety function ($\theta(.)$-scale). We first introduce the cardinality-level benchmark and the formal definition of parameter Δ:

Algorithm 4: Grow-and-Tune Local Search

Input: Ground set $\mathcal{T}$ with $n := |\mathcal{T}|$, accuracy parameter $\delta > 0$, GS parameter γ, tuning threshold $\eta := \frac{\delta\gamma}{8n}$

Output: A subset $\widehat{S} \subseteq \mathcal{T}$

$S_0 \leftarrow \emptyset$

for $\ell \leftarrow 1$ **to** n **do**

  Set the level-ℓ sample count $m_\ell \leftarrow \frac{8\ell^2}{\delta^2\gamma^2} \ln(2n^5[(n-\ell+1) + \ell(n-\ell)(1+\frac{1}{\eta})])$

  `// Grow to cardinality` ℓ

  $C_\ell^{\text{grow}} \leftarrow \mathcal{G}(S_{\ell-1})$

  Estimate $\widehat{\theta}(S)$ for every $S \in C_\ell^{\text{grow}}$ using m_ℓ samples

  $S_\ell \leftarrow \arg\max_{S \in C_\ell^{\text{grow}}} \widehat{\theta}(S)$

  `// Tune within cardinality` ℓ

  while $\mathcal{N}(S_\ell) \neq \emptyset$ **do**

    Estimate $\widehat{\theta}(S')$ for every $S' \in \mathcal{N}(S_\ell)$ using m_ℓ samples

    $S^{\text{cand}} \leftarrow \arg\max_{S' \in \mathcal{N}(S_\ell)} \widehat{\theta}(S')$

    if $\widehat{\theta}(S^{\text{cand}}) > \widehat{\theta}(S_\ell) + \eta$ **then**

      $S_\ell \leftarrow S^{\text{cand}}$

    else

      break

$\widehat{S} \leftarrow \arg\max_{\ell \in [n]} \ell\, \widehat{\theta}(S_\ell)$

return $\widehat{S}$

Fix a level $\ell \in \{1, \ldots, n\}$ and let $S_\ell^\star \in \arg\max_{S \subseteq \mathcal{T}:\ |S|=\ell} \theta(S)$ be an optimal set for the level ℓ. This is equivalently a minimizer of $R(S)$ among all $|S| = \ell$ sets. By Lemma 4.1, we fix a nested chain of level-optimal sets $S_\ell^\star \subset S_{\ell+1}^\star$ for all ℓ. Let $S^\star$ be the globally optimal set as $S^\star$ is the set (among the $S_\ell^\star$) that maximizes $\ell \cdot \theta(S_\ell^\star)$ and let $\ell^\star = |S^\star|$. We now state the local GS scale used in the analysis. If we recall in Def. 2.2, the GS-property is active only in a small interval of Δ, we use the following assumption of Δ in our setting. Define $\bar{r} := \max_{a \in S^\star} r_a$, and $\Delta^\star := 2\bar{r}$. Throughout the analysis, we assume that f satisfies the Gradient-Smoothness property with parameters $(\Delta^\star, \gamma)$. Since $S_\ell^\star \subseteq S^\star$ for every $\ell \leq \ell^\star$, the GS property is active in cardinality levels $\ell \leq \ell^\star$ on an interval of length at most $2\bar{r}$. Our main guarantee is the following:

Theorem 4.2. *Let $S^\star \subseteq \mathcal{T}$ be the optimal set and let $\epsilon > 0$. Suppose f satisfies the GS property with paramters $(\Delta^\star, \gamma)$, where $\Delta^\star := 2\max_{a \in S^\star} r_a$. Run algorithm 4 with $\delta := \frac{\epsilon}{8n^2}$ and $\eta = \frac{\delta\gamma}{8n}$. Then with probability $\geq 1 - \frac{1}{n^4}$, the output $\widehat{S}$ of Algorithm 4 satisfies the condition:*

$$Obj(\widehat{S}) \geq (Obj(S^\star) - \epsilon/4).$$

We begin our analysis with a simple averaging lemma showing that if any size-ℓ set B has larger accumulated risk than a size-ℓ set A, then a single swap in set B can reduce the risk by $1/\ell$-fraction of the gap.

Lemma 4.3. *Suppose A and B are two subsets of $\mathcal{T}$ and $|A| = |B| = \ell$. Suppose $R(A) < R(B)$. Then there exist $a \in A$ and $b \in B$ such that after swapping out b for a, i.e., taking $B' = B \setminus \{b\} \cup \{a\}$, we have*

$$R(B') \leq R(B) - \frac{(R(B) - R(A))}{\ell}.$$

Proof Overview of Lemma 4.3. Since A and B have the same cardinality but $R(A) < R(B)$, the elements in $B \setminus A$ have greater total risk than those in $A \setminus B$. By averaging, removing the highest-risk element of $B \setminus A$ and adding the lowest-risk element of $A \setminus B$ decreases the risk by at least $(R(B) - R(A))/\ell$. This gives the desired one-swap neighbor. The complete proof is given in Appendix D.2. □

The next lemma gives the uniform concentration event used throughout the analysis.

Lemma 4.4. *Let $\delta > 0$ be given. Consider the Algorithm 4 under Bernoulli subset-level feedback model adaptively queries $\theta(S)$ values for a sequence of sets $S_1, S_2, \ldots, S_N$. (Formally, at any q, the choice of S_t depends on $S_1, S_2, \ldots,$ as well as the estimates $\widehat{\theta}(S_i)$, for $i < q$.) Then if each query is evaluated using $\frac{8\ell^2}{\delta^2\gamma^2} \ln(2n^5[(n-\ell+1) + \ell(n-\ell)(1+\frac{1}{\eta})])$ samples, with probability at least $1 - \frac{1}{n^4}$, we have $|\theta(S_q) - \widehat{\theta}(S_q)| < \frac{\delta\gamma}{4\ell}$ for* all $1 \le q \le N$.

Let $\mathcal{E}$ denote the uniform-concentration event. (See Appendix A.3 for the proof.)

We now use the Lemma 4.3 together with uniform estimation Lemma 4.4 and the GS implication (Lemma 2.3) to prove the main inductive invariant controlling the quality of the set maintained at the end of each cardinality level.

Lemma 4.5. *Let S_ℓ be the final representative at level ℓ in Algorithm 4. For all $0 \le \ell \le |S^\star|$, we have*

$$\theta(S_\ell) \ge \theta(S_\ell^\star) - \ell\delta.$$

Proof Overview of Lemma 4.5: We prove the claim by induction on ℓ, conditioning on the concentration event of Lemma 4.4. If $\theta(S_\ell) \ge \theta(S_{\ell-1}) - \delta$, the claim follows directly from the induction hypothesis and the nested structure of the level-optimal sets (Lemma 4.1).

Otherwise, the first step is to show that the Grow procedure does not take us too far in terms of aggregate risk. Let a be a minimum-risk element of $S^\star \setminus S_{\ell-1}$. By comparing S_ℓ with the grow candidate $S_{\ell-1} \cup \{a\}$, and using Lemma 4.4 together with GS, we obtain $R(S_\ell) \le R(S_{\ell-1}) + 2r_a$. This is important because $2r_a \le \Delta^\star$, allowing us to use the local GS property.

Now, when the Tune procedure terminates, no one-swap neighbor can improve the empirical safety probability by more than η. Using Lemma 4.4 again, this gives $\theta(S') \le \theta(S_\ell) + \frac{5\delta\gamma}{8\ell}$, where $S' \in \mathcal{N}(S_\ell)$. To finish, suppose that $\theta(S_\ell^\star) > \theta(S_\ell) + \ell\delta$. We consider the relative risks of $S_\ell^\star$, $S_{\ell-1}$, and S_ℓ. The averaging argument of Lemma 4.3, together with the GS implication of Lemma 2.3, is used to obtain a one-swap improvement exceeding the above bound, giving the desired contradiction. The complete case analysis is given in Appendix D.3.

We now joint these arguments to prove Theorem 4.2.

Proof Overview of Theorem 4.2: Condition on the uniform concentration event of Lemma 4.4. Applying Lemma 4.5 at the optimal cardinality $\ell^\star$, we obtain $\mathrm{Obj}(S_{\ell^\star}) \ge \mathrm{OPT} - {\ell^\star}^2\delta$. Now, Algorithm 4 returns the representative with the largest empirical objective. By Lemma 4.4, this final selection loses at most an additional $\delta\gamma/2$. Therefore,

$$\mathrm{Obj}(\widehat{S}) \ge \mathrm{OPT} - {\ell^\star}^2\delta - \frac{\delta\gamma}{2} \ge \mathrm{OPT} - \frac{\epsilon}{4},$$

where the last inequality follows from $\delta = \epsilon/(8n^2)$. The complete proof is given in Appendix D.4.

4.4 Minimizing Regret: Optimistic Grow-and-Tune

A direct way to turn Algorithm 4 into an online algorithm is by using a simple explore-then-commit approach, which turns Theorem 4.2 into a sublinear regret algorithm that achieves $\widetilde{O}(T^{3/4})$ regret. Informally (and ignoring the problem parameters), the Grow-and-Tune algorithm achieves an error

of ε after roughly $\widetilde{O}(\frac{1}{\varepsilon^3})$ "exploration" steps. Each step incurs a regret of a constant, and thus the total regret if run for T steps would be $\varepsilon T + \frac{1}{\varepsilon^3}$. Balancing ε yields the regret bound to be $\widetilde{O}(T^{3/4})$.

To obtain a regret guarantee of $\widetilde{O}(\sqrt{T})$, in this section, we develop an analog of the Grow-and-Tune algorithm that is inspired by the classic UCB algorithm (e.g., see [7]). Two major difficulties arise. Firstly, the algorithm searches across different cardinality levels, and not all of them are equally promising. For example, earlier levels may have objective far below OPT simply because their cardinality $|S|$ is small which is the throughput factor in the objective. Repeatedly sampling such a cardinality level would therefore incur substantial regret. At the same time, these smaller cardinality levels cannot simply be ignored: the Grow step at cardinality ℓ starts from the representative constructed at cardinality $\ell - 1$. This implies that there is dependence across cardinality levels where the reliability of the representative set in the later levels depends on the correctness of the earlier levels and we can observe this in the proof of Lemma 4.5. Secondly, UCB is traditionally used with a fixed set of arms. Since our search space is exponential, we need to carefully incorporate UCB ideas into our local search procedure.

We now present Algorithm 5 titled *Optimistic Grow-and-Tune* that addresses these challenges. It works in two stages: In the first part, it runs Algorithm 4 once *at a coarse accuracy* to obtain one initial representative S_ℓ^0 for each cardinality level ℓ. In the second part, it exploits some local properties to refine the representatives through online sampling. The purpose of this warm start is not to solve every level accurately, but to get *good-enough* representative for each level *without wasting much samples*. In section 4.4.1 and 4.4.2, we show that warm-start procedure produces a coarse representative S_ℓ^0 for each cardinality level ℓ, with accuracy sufficient for subsequent online refinement. Crucially, for the cardinality level ℓ that contains the optimal set $S^\star$, the representative S_ℓ^0 lies within the Gradient-Smoothness window $\Delta^\star$ of the optimal set $S^\star$. Once within the GS-window, we establish two more interesting local properties: firstly, even if the current representative S_ℓ is still far from the best set of the same cardinality, there are some one-swap neighbor set which ensures noticeable improvement. And secondly, these "promising" one-swap neighbors helps us to upper-bound the objective of the *best-set* of that respective cardinality level ℓ. We exploit these two properties for the online sampling round. We refine the representatives sets following the first property. However, we don't uniformly refine all the cardinality levels, rather, only refine the promising ones that may contain the optimal set $S^\star$. We keep *optimistic upper-bound index* for each cardinality level following the second property stated above and dynamically spend samples on the level with the largest optimistic-index in each round. This design strategy helps to avoid wasting samples on suboptimal levels. We present the formal algorithm with details in Section 4.4.3 and the regret analysis in Section 4.4.4.

4.4.1 Setup and Warm Start. Recall from Lemma 4.1 that the cardinality-level optimal sets can be chosen to form a nested chain: $S_0^\star \subset S_1^\star \subset \cdots \subset S_n^\star$, where $S_\ell^\star \in \arg\max_{S\subseteq\mathcal{T},|S|=\ell} \theta(S)$. Thus, for each *cardinality-level optimal set* $S_\ell^\star$ of ℓ, we define $\theta_\ell^\star := \theta(S_\ell^\star)$ and $V_\ell^\star := \ell\theta_\ell^\star$. We also define the notations of *global-optimal set* as follows: $\ell^\star \in \arg\max_{\ell\in[n]} V_\ell^\star$, $S^\star := S_{\ell^\star}^\star$, and $\mathrm{OPT} := V_{\ell^\star}^\star$.

For any size-ℓ set S, we denote its objective gap from the level-ℓ optimal set as: $G_\ell(S) := V_\ell^\star - \ell\theta(S)$. As in Section 4.3, define the local GS scale around the global optimum by $\Delta^\star := 2\max_{a\in S^\star} r_a$, and assume that f satisfies the Gradient-Smoothness property with parameter γ on intervals of length at most $\Delta^\star$. The algorithm does not need to know $\Delta^\star$, but we assume that γ is known up to constant factors. We also use the singleton scale $g := \max_{i\in\mathcal{T}} \theta(\{i\})$, similar to the Section 3.4, and assume that g is known upto a constant factor.

Warm Start. The first component of Algorithm 5 is a warm-start procedure that initializes one representative set S_ℓ^0 for each level ℓ. The goal is to obtain locally good representatives, crucially, to

ensure that the initial representative S_ℓ^0 are *somewhat close-enough* to its corresponding cardinality level optimum $S_\ell^\star$. Therefore, in the *warm-start*, we run our old Algorithm 4 with the coarse parameter, $\delta_{\text{init}} := \frac{\gamma g}{128n^3}$, and $\eta_{\text{init}} := \frac{\gamma\delta_{\text{init}}}{8n}$.

Let $S_1^0, S_2^0, \ldots, S_n^0$ be the representatives produced by this warm-start run. Let $\mathcal{E}_{\text{ws}}$ denote the warm-start concentration event of Lemma 4.4. By Lemma 4.5, on $\mathcal{E}_{\text{ws}}$, for every $\ell \le \ell^\star$,

$$\theta(S_\ell^0) \ge \theta(S_\ell^\star) - \ell\delta_{\text{init}}. \tag{2}$$

Consequently, the level-ℓ objective gap satisfies

$$G_\ell(S_\ell^0) := V_\ell^\star - \ell\theta(S_\ell^0) \le \ell^2\delta_{\text{init}}. \tag{3}$$

Thus the warm start gives a coarse representative at each relevant cardinality level and guarantees in terms of success probability $\theta(.)$ (in eq. (2)), whereas the GS property applies only in the accumulated risk scale $R(.)$. Therefore, we next establish guarantees in the accumulated risk scale.

4.4.2 From Local Search to an Optimistic Level Index. In this section we establish some important local properties of the initial sets S_ℓ^0 in the risk scale $R(.)$. We first present a strong guarantee for the optimal cardinality level $\ell^\star$ that its initial set $S_{\ell^\star}^0$ is within $\Delta^\star$-window of its cardinality level optimal set $S_{\ell^\star}^\star = S^\star$. Therefore we can exploit GS properties here. This consequently helps us to guarantee another local property: by *one-swap local moves*, we can get better representative set S_ℓ for each level ℓ.

For a level ℓ, let $\bar{r}_\ell := \max_{a\in S_\ell^\star} r_a$. We call a size-$\ell$ set S ℓ-window-good if

$$R(S_\ell^\star) \le R(S) \le R(S_\ell^\star) + 2\bar{r}_\ell. \tag{4}$$

The left inequality holds automatically because $S_\ell^\star$ minimizes accumulated risk among all size-ℓ sets. For the globally optimal level, $2\bar{r}_{\ell^\star} = 2\max_{a\in S^\star} r_a = \Delta^\star$.

Lemma 4.6. *Condition on $\mathcal{E}_{\text{ws}}$. The warm-start representative of the globally optimal cardinality-level, $S_{\ell^\star}^0$ is $\ell^\star$-window good, that is, $R(S_{\ell^\star}^\star) \le R(S_{\ell^\star}^0) \le R(S_{\ell^\star}^\star) + 2\bar{r}_{\ell^\star}$, where, $\bar{r}_{\ell^\star} := \max_{a\in S_{\ell^\star}^\star} r_a$. Equivalently, since $S^\star = S_{\ell^\star}^\star$ and $\Delta^\star = 2\max_{a\in S^\star} r_a = 2\bar{r}_{\ell^\star}$, we have*

$$R(S^\star) \le R(S_{\ell^\star}^0) \le R(S^\star) + \Delta^\star.$$

Proof Overview of Lemma 4.6: The main challenge is to translate the warm-start guarantee in (2), which is in terms of success probability, into a guarantee on the aggregate risk. To this end, for each $\ell < n$, define the outside-window gap $\omega_\ell := \theta(S_\ell^\star) - \max_{|S|=\ell, R(S)>R(S_\ell^\star)+2\bar{r}_\ell} \theta(S)$, where the maximum over an empty set is $-\infty$. Thus, ω_ℓ measures the loss in success probability of the best size-ℓ set outside the GS window. The key observation is that if ω_ℓ is sufficiently small, then level ℓ cannot be globally optimal. Indeed, using the nested structure of Lemma 4.1 and the GS property (Definition 2.2), we obtain $\theta_\ell^\star - \theta_{\ell+1}^\star \le \left(1 + \frac{1}{\gamma}\right)\omega_\ell$. The argument compares the drops of f over neighboring risk intervals of total length at most $2\bar{r}_\ell$. Consequently, whenever $V_\ell^\star \ge g/2$, an outside-window gap $\omega_\ell < \gamma g/(4n^2)$ implies $V_{\ell+1}^\star > V_\ell^\star$.

Now suppose $\ell^\star < n$ and the warm-start set $S_{\ell^\star}^0$ is not $\ell^\star$-window-good. By the definition of $\omega_{\ell^\star}$ and the warm-start guarantee (2), obtained from Lemma 4.5, we have $\omega_{\ell^\star} \le \theta_{\ell^\star}^\star - \theta(S_{\ell^\star}^0) \le \ell^\star\delta_{\text{init}} \le \frac{\gamma g}{128n^2}$. Moreover, by Lemma 3.7, $V_{\ell^\star}^\star = \text{OPT} \ge g$. The preceding observation would therefore give $V_{\ell^\star+1}^\star > V_{\ell^\star}^\star$, contradicting the optimality of $\ell^\star$. For $\ell^\star = n$, the claim is immediate since the only size-n set is the ground set itself. The complete argument is given in Appendix E.1.

We now use this property to relate the quality of a representative to its one-swap neighborhood. In particular, if a window-good representative is still far from its cardinality-level optimum", then at least one neighboring set must offer a substantial improvement.

Lemma 4.7. *Fix a level ℓ and a size-ℓ set S, where S is ℓ-window good. Under the active Gradient-Smoothness condition on the interval between $S_\ell^\star$ and S,*

$$G_\ell(S) \le \frac{\ell^2}{\gamma} \max_{S' \in \mathcal{N}(S)} [\theta(S') - \theta(S)]_+,$$

where $\mathcal{N}(S) := \{(S \setminus \{i\}) \cup \{j\} : i \in S,\ j \notin S\}$ is the single-swap tune neighborhood.

Proof Overview of Lemma 4.7: By Lemma 4.3, some one-swap neighbor S' reduces the aggregate-risk gap to $S_\ell^\star$ by at least a $1/\ell$ fraction. Since S is ℓ-window-good, Lemma 2.3 converts this risk reduction into a safety-probability improvement of at least $(\gamma/\ell^2)G_\ell(S)$. Taking the best one-swap neighbor yields the claimed bound. The complete proof is given in Appendix E.1.

Lemma 4.7 gives us a way to bound the remaining cardinality-level gap using local improvements. The difficulty is that these improvements are not known exactly under Bernoulli feedback. We therefore use the following confidence bounds:

Confidence Bounds: Recall that $g = \max_{a \in \mathcal{T}} \theta(\{a\})$. Then every non-empty set A satisfies $\theta(A) \le g$. Let $N_t(A)$ be the number of the samples of A before round t, $\widehat{\theta}_t(A)$ be its empirical mean and $L_t := c \log(nT)$ where $c > 0$. For $N_t(A) \ge 1$, we define singleton-scale Bernstein confidence radius

$$b_t(A) := c\left(\sqrt{\frac{gL_T}{N_t(A)}} + \frac{L_T}{N_t(A)}\right)$$

. For $N_t(A) = 0$, set $b_t(A) = g$. We also define Upper and Lower confidence bound of set A as: $\mathrm{UCB}_t(A) := \min\{g, \widehat{\theta}_t(A) + b_t(A)\}$, and $\mathrm{LCB}_t(A) := \max\{0, \widehat{\theta}_t(A) - b_t(A)\}$.

For an unqueried set, the same convention gives $\mathrm{UCB}_t(A) = g$ and $\mathrm{LCB}_t(A) = 0$. The following standard concentration lemma ensures that the intervals are valid throughout the online stage.

Lemma 4.8. *With probability at least $1 - T^{-2}$, every set A queried by the algorithm and every time $t \le T$ at which its estimate is used satisfy $\mathrm{LCB}_t(A) \le \theta(A) \le \mathrm{UCB}_t(A)$, and $\mathrm{UCB}_t(A) - \mathrm{LCB}_t(A) \le 2b_t(A)$.*

We denote this online confidence event by $\mathcal{E}_{\mathrm{on}}$. For the remainder of the analysis, let $\mathcal{E} := \mathcal{E}_{\mathrm{ws}} \cap \mathcal{E}_{\mathrm{on}}$.

The proof follows from Bernstein's inequality and a union bound; see Appendix E.2.

We now define the local cardinality-level quantities used in the algorithm and the analysis:

Local cardinality-level quantities: Fix a cardinallity level $\ell \in [n]$ and its size-ℓ representative S_ℓ, then its one-swap neighborhood

$$\mathcal{N}(S_\ell) := \{(S_\ell \setminus \{i\}) \cup \{j\} : i \in S_\ell,\ j \notin S_\ell\} \tag{5}$$

Among the one-swap neighbors of S_ℓ, let

$$A_\ell^+(t) \in \arg\max_{A \in \mathcal{N}(S_\ell)} \mathrm{UCB}_t(A) \tag{6}$$

be its most optimistic neighbor, and define

$$I_\ell^+(t) := \left[\mathrm{UCB}_t(A_\ell^+(t)) - \mathrm{LCB}_t(S_\ell)\right]_+. \tag{7}$$

On $\mathcal{E}$, $\max_{A \in \mathcal{N}(S_\ell)} [\theta(A) - \theta(S_\ell)]_+ \le I_\ell^+(t)$. Thus, whenever S_ℓ is window-good, Lemma 4.7 gives $G_\ell(S_\ell) \le \frac{\ell^2}{\gamma} I_\ell^+(t)$. This motivates the optimistic level index

$$U_\ell(t) := \ell\, \mathrm{UCB}_t(S_\ell) + \frac{\ell^2}{\gamma} I_\ell^+(t). \tag{8}$$

This formula (8) now has a completely transparent interpretation: the first part $\ell\,\mathrm{UCB}_t(S_\ell)$ captures the optimistic value of the current representative whereas the second part $\frac{\ell^2}{\gamma} I_\ell^+(t)$ expresses optimistic bound on what the cardinality-level may still be missing.

We also need a lower-confidence certificate for an improving swap. Let the certified "most guaranteed" neighbor $A_\ell^-(t) \in \arg\max_{A \in \mathcal{N}(S_\ell)} \mathrm{LCB}_t(A)$, and define

$$D_\ell^-(t) := \left[\mathrm{LCB}_t(A_\ell^-(t)) - \mathrm{UCB}_t(S_\ell)\right]_+. \tag{9}$$

Hence $D_\ell^-(t) > 0$ certifies that $A_\ell^-(t)$ is truly better than the current representative. For $\ell = n$, the one-swap neighborhood is empty; we use the conventions $A_n^+(t) = S_n$ and $I_n^+(t) = D_n^-(t) = 0$.

4.4.3 Blocked Optimistic Grow-and-Tune. We now present the formal online part of the algorithm. Algorithm 5 contains the main sampling loop, and Algorithm 7 contains its two helper subroutines. We first introduce the concept of **blocks**. Intuitively, we already know that, during the online round, for each cardinality level ℓ, the algorithm keeps a representative set S_ℓ and it refines the representative by collecting more samples. We organize these refinement rounds within each cardinality-level as *blocks*. For a level ℓ and block k, the algorithm has a current representative S_ℓ and a "target" cardinality-level obtimality gap $\varepsilon_{\ell,k}$ (upper bound of the objective gap between S_ℓ and $S_\ell^\star$). The algorithm collects more sample to certify that the current representative S_ℓ or any other one-swap neighbor of S_ℓ maintains this gap with $S_\ell^\star$. Once the gap is certified, the algorithm moves to next block $k+1$ with a *geometrically-reduced* target gap $\varepsilon_{\ell,k+1}$. Thus, the online stage progressively reduces an upper bound on the optimality gap at each cardinality level.

Formally, For every ℓ, we initialize that upper bound on the cardinality-level optimality gap or *block-scale* as

$$\varepsilon_{\ell,0} := \ell g. \tag{10}$$

This is a valid universal bound because $0 \le G_\ell(S) \le V_\ell^\star \le \ell g$ for every size-ℓ set S. This also resolves the issue we discussed earlier: although the warm start gives the much sharper bound $G_\ell(S_\ell^0) \le \ell^2 \delta_{\mathrm{init}}$ for $\ell \le \ell^\star$, the algorithm does not know which levels satisfy this condition. We therefore initialize every level with the universal scale ℓg; the sharper warm-start bound is used in the analysis of the globally optimal level $\ell^\star$.

We also define

$$\rho_\ell := 1 - \frac{\gamma}{8\ell}, \qquad \varepsilon_{\ell,k+1} := \rho_\ell \varepsilon_{\ell,k}. \tag{11}$$

Each level progresses through these geometrically decreasing blocks. Within a block k, its representative S_ℓ remains fixed while the algorithm collects evidence until one of two events can be certified: (1) If $D_\ell^-(t) \ge \frac{\gamma}{8\ell^2}\varepsilon_{\ell,k}$, a one-swap neighbor is certified to improve the representative sufficiently, and we replace S_ℓ by $A_\ell^-(t)$. (2) Alternatively, if $I_\ell^+(t) \le \frac{\gamma}{\ell^2}\varepsilon_{\ell,k+1}$, then Lemma 4.7 certifies that the cardinality-level gap is already at most the next target scale.

In either case, the block closes and its target is reduced from $\varepsilon_{\ell,k}$ to $\varepsilon_{\ell,k+1}$. We stop refining a level once its target drops below $\varepsilon_{\min} := \frac{g}{T}$. Such a level is called **frozen**. For a frozen level we use $U_\ell(t) = \ell\,\mathrm{UCB}_t(S_\ell) + \varepsilon_{\min}$.

Algorithm 5 summarizes the complete procedure. Algorithm 5 uses two helper subroutines. ComputeIndex(ℓ) updates the optimistic index U_ℓ and the local confidence quantities using the current observations. ClosePhase(ℓ) checks the two conditions described above. If a swap is certified, it updates the representative S_ℓ; otherwise, if the remaining improvement is sufficiently small, it keeps S_ℓ unchanged. In either case, the block target is reduced geometrically, and the procedure repeats until no further block can close or the level becomes frozen. Neither subroutine collects new samples. Algorithm 5 calls ClosePhase for all levels initially and for the selected level after each online observation. Here ε_ℓ denotes the *current block target* $\varepsilon_{\ell,k}$. Thus, whenever a block

closes and the algorithm updates $\varepsilon_\ell \leftarrow \rho_\ell \varepsilon_\ell$, this corresponds to moving from $\varepsilon_{\ell,k}$ to $\varepsilon_{\ell,k+1} = \rho_\ell \varepsilon_{\ell,k}$. The complete helper pseudocode appears in Appendix E.3.

Algorithm 5: Optimistic Grow-and-Tune

Input: Ground set $\mathcal{T}$ with $n := |\mathcal{T}|$, horizon T, GS parameter γ, and known singleton scale $g = \max_{a\in\mathcal{T}} \theta(\{a\})$
Output: Played sets $Q_1, \ldots, Q_T$

```
// Warm start
```
Set $\delta_{\text{init}} \leftarrow \frac{\gamma g}{128n^3}$ and $\eta_{\text{init}} \leftarrow \frac{\gamma\delta_{\text{init}}}{8n}$; run Algorithm 4 to obtain $S_1^0, \ldots, S_n^0$;
```
// Online Rounds
```
Set $L_T \leftarrow c\log(nT)$, $\varepsilon_{\min} \leftarrow \frac{g}{T}$.
For each level ℓ, initialize $S_\ell \leftarrow S_\ell^0$, $\varepsilon_\ell \leftarrow \ell g$, $\rho_\ell \leftarrow 1 - \frac{\gamma}{8\ell}$, $\text{frozen}_\ell \leftarrow \text{false}$.
For every queried set A, maintain $N(A)$ and $\widehat{\theta}(A)$.
```
// Initial closure: make every level stable before the first pull
```
for $\ell = 1, \ldots, n$ **do**
  `ClosePhase(`ℓ`);`

for $t = 1, \ldots, T$ **do**
  for $\ell = 1, \ldots, n$ **do**
    `ComputeIndex(`ℓ`);`
  Choose $\ell_t \in \arg\max_{\ell\in[n]} U_\ell$.
  if $\text{frozen}_{\ell_t} = \text{true}$ **then**
    Set $Q_t \leftarrow S_{\ell_t}$
  else
    Set $Q_t \leftarrow \arg\max_{Q\in\{S_{\ell_t}, A^+_{\ell_t}\}} b_t(Q)$.
  Play Q_t, observe $Y_t(Q_t) \in \{0,1\}$, Update $\widehat{\theta}(Q_t)$ and $N(Q_t)$;
  ```
  // Step to keep all active level stable before the next pull:
  ```
  `ClosePhase(`ℓ_t`);`

Now, we present the key invariant. We show how the optimistic index $U_{\ell^\star}(t)$ of the optimal level $\ell^\star$ never falls below the OPT.

Lemma 4.9. *Condition on $\mathcal{E}$. At every time t, the representative at the optimal level $S_{\ell^\star}$ is $\ell^\star$-window-good and satisfies $G_{\ell^\star}(S_{\ell^\star}) \le \varepsilon_{\ell^\star,k_{\ell^\star}}$. Moreover, at every online round, $U_{\ell^\star}(t) \ge \text{OPT}$.*

Proof Overview of Lemma 4.9: By Lemma 4.6, the warm-start representative $S^0_{\ell^\star}$ is $\ell^\star$-window-good. Moreover, $G_{\ell^\star}(S^0_{\ell^\star}) \le \ell^\star g = \varepsilon_{\ell^\star,0}$, so the invariant holds initially.

We now show that it is preserved whenever a block at level $\ell^\star$ closes. If the block closes without a swap, the no-swap certificate, together with Lemmas 4.7 and 4.8, gives $G_{\ell^\star}(S_{\ell^\star}) \le \varepsilon^+$, where $\varepsilon^+ = (1 - \gamma/(8\ell^\star))\varepsilon$. Otherwise, the swap certificate guarantees a true improvement of at least $\gamma\varepsilon/(8\ell^{\star 2})$ in safety probability. Thus, the objective gap decreases by at least $\gamma\varepsilon/(8\ell^\star)$, and is again at most ε^+. Moreover, since f is strictly decreasing, the accepted swap reduces the aggregate risk, thereby preserving window-goodness.

Finally, when level $\ell^\star$ is non-frozen, Lemmas 4.7 and 4.8 guarantee that its optimistic index satisfies $U_{\ell^\star}(t) \ge \text{OPT}$. Once the level is frozen, the maintained gap bound $G_{\ell^\star}(S_{\ell^\star}) \le \varepsilon_{\min}$ gives the same conclusion through the frozen index. The complete proof is given in Appendix E.4.

4.4.4 *Regret Analysis.*

At online round t, Algorithm 5 chooses $\ell_t \in \arg\max_{\ell\in[n]} U_\ell(t)$.

Lemma 4.9 ensures that, $U_{\ell_t}(t) \geq U_{\ell^\star}(t) \geq \mathrm{OPT}$. Thus the optimistic index of the selected cardinality-level ℓ_t always upper-bounds the optimistic index of global optimum $\ell^\star$. If Q_t denotes the set sampled at round t, we define the instantaneous regret at round t as

$$r_t := \mathrm{OPT} - |Q_t|\theta(Q_t).$$

The significant regret is accumulated while a block remains unresolved (non-frozen) cause in a frozen round the block scale becomes tiny as $\varepsilon_{min} = \frac{g}{T}$ incurring very small regret. On the other hand, in an unresolved block, neither certificate $D^-_\ell(t) \geq \frac{\gamma}{8\ell^2}\varepsilon_{\ell,k}$ nor $I^+_\ell(t) \leq \frac{\gamma}{\ell^2}\varepsilon_{\ell,k+1}$ holds, which forces the respective confidence intervals to remain wide. Consequently, regret is also charged directly to confidence width.

Lemma 4.10. *Fix a non-frozen block k of level ℓ. Let $N_{\ell,k}$ be the number of sampling rounds spend in the block, and let $H_\ell := 1 + |N(S_\ell)| \leq 1 + \ell(n-\ell) \leq n^2$ denotes the size of its local family. Then on $\mathcal{E}$, the accumulated regret in this block is $\mathcal{R}_{\ell,k} \leq \widetilde{O}(\frac{\ell^2}{\gamma}|\sqrt{gH_\ell N_{\ell,k}} + H_\ell|)$.*

Proof Overview of Lemma 4.10: Consider an unresolved block at level ℓ, and let $S = S_\ell$ and $A = A^+_\ell(t)$. Since Algorithm 5 closes all possible blocks before sampling, neither closing condition holds. Combining these conditions with Lemma 4.8 gives $b_t(S) + b_t(A) \geq c(\gamma/\ell^2)\varepsilon_{\ell,k}$ for some constant $c > 0$, and bounds $I^+_\ell(t)$ in terms of these confidence widths.

By Lemma 4.9, the selected level satisfies $U_\ell(t) \geq \mathrm{OPT}$. Since the algorithm samples the more uncertain set between S and A, we obtain $r_t \leq C(\ell^2/\gamma)b_t(Q_t)$.

Finally, the representative and its neighborhood remain fixed within a block, so all samples belong to a family of size H_ℓ. Summing their Bernstein radii gives $\sum_{t \text{ in block}} b_t(Q_t) = \widetilde{O}(\sqrt{gH_\ell N_{\ell,k}} + H_\ell)$, proving the claimed bound. The complete proof is in Appendix E.5.

Thus the final regret of Algorithm 5 becomes the aggregated value of the warm-start, non-frozen and frozen regrets as the following theorem:

Theorem 4.11. *Assume $g = \max_{a\in\mathcal{T}}\theta(\{a\})$ is known. Let $S^\star$ be an optimal set, $\ell^\star = |S^\star|$, and define $\Delta^\star := 2\max_{a\in S^\star} r_a$. Assume f satisfies the GS property with parameter γ on every interval of length at most $\Delta^\star$. Then the regret of the blocked optimistic Grow-and-Tune algorithm satisfies $\mathbb{E}[\mathcal{R}(T)] \leq \widetilde{O}\Big(\frac{n^4}{\gamma^{3/2}}\sqrt{gT} + \frac{n^6}{\gamma^2} + \frac{n^{16}}{\gamma^6 g^2}\Big)$. In particular, since $g \leq 1$, and the considering instance-dependent additive term $\mathcal{A}_{\mathrm{inst}}(n,\gamma,g) := \widetilde{O}(\frac{n^6}{\gamma^2} + \frac{n^{16}}{\gamma^6 g^2})$, the regret bound becomes, $\mathbb{E}[\mathcal{R}(T)] \leq \widetilde{O}\Big(\frac{n^4}{\gamma^{3/2}}\sqrt{T}\Big)$.*

Proof Overview of Theorem 4.11: We decompose the regret into warm-start, non-frozen, and frozen rounds. The sample complexity of Algorithm 4, together with $\mathrm{OPT} \leq ng$, bounds the warm-start contribution by $\widetilde{O}(n^{16}/(\gamma^6 g^2))$. For non-frozen rounds, each level has $\widetilde{O}(\ell/\gamma)$ geometrically decreasing blocks; summing Lemma 4.10 gives $\widetilde{O}(n^4\sqrt{gT}/\gamma^{3/2} + n^6/\gamma^2)$. Finally, optimism at frozen levels (Lemma 4.9) bounds their regret by confidence widths and a residual g/T per round. Details, including failure events, are given in Appendix E.6.

5 Conclusion

We studied subset selection under binary stochastic feasibility feedback with hidden additive risks. Under monotonicity alone, Swap-and-Grow achieves a $t/(t+1)$-approximation; an exact-oracle lower bound shows that beating this factor requires $n^{\Omega(t)}$ queries. Under strict monotonicity and local Gradient-Smoothness, Grow-and-Tune achieves arbitrarily small additive error. Both algorithms extend to online learning with $\widetilde{O}(\sqrt{T})$ regret against their respective benchmarks. These results demonstrate the value of additive structure under restricted subset-level feedback.

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A Concentration Lemmas and Their Deferred Proofs

A.1 Adaptive Subset Evaluation

Lemma A.1. *Suppose an adaptive algorithm generates a sequence of subset evaluations indexed by $S_1, S_2, ..., S_N$, where, at step q it takes a set S_q based on the history upto step $q-1$. Then it draws m_q samples iid Bernoulli of $\theta(S_q)$ and found an empirical mean of $\widehat{\theta}(S_q)$ with an tolerance of $\epsilon_q > 0$, then,*

$$\Pr(\forall q \in [N] : |\widehat{\theta}(S_q) - \theta(S_q)| \leq \epsilon_q) \geq 1 - \sum_{q=1}^{N} 2e^{-2m_q\epsilon_q^2}$$

Proof. We define the bad event that the estimation fail at iteration q as:

$$B_q := \{|\widehat{\theta}(S_q) - \theta(S_q)| > \epsilon_q\}$$

We also define that till iteration q, the cumulative good event H_q by:

$$H_q := \cap_{j=1}^{q} B_j^c$$

Now, let G_{q-1} denote the history upto step $q-1$ that include all queried sets and observed sample by the adaptive algorithm before selecting S_q.

Now, conditioning on G_{q-1}, the set S_q become fized and if we use m_q samples to estimate it, then by hoeffding inequality,

$$\Pr(B_q|G_{q-1}) \leq 2e^{-2m_q\epsilon_q^2}. \tag{12}$$

Now, we derive H_q using induction.For the base case $q = 1$, since $H_1 = B_1^c$, we have: $\Pr(H_1) = Pr(B_1^c) \geq 1 - 2e^{-2m_1\epsilon_1^2}$, again by Hoeffding inequality.

Now, fix any $q \geq 2$. Since H_{q-1} determined by first $q-1$ steps,

$$\Pr(H_q) = \Pr(H_{q-1} \cap B_q^c) = \Pr(B_q^c|H_{q-1}) \Pr(H_{q-1}).$$

Now, event H_{q-1} depends on previous history till $q-1$ and encoded as G_{q-1}. thus, using eq. (12), we get,

$$\Pr(H_q) =\geq (1 - 2e^{-2m_q\epsilon_q^2}) \cdot \Pr(H_{q-1}).$$

Applying this recursively from q down to 1, we obtain:

$$\Pr(H_q) \geq \prod_{j=1}^{q} (1 - 2e^{-2m_j\epsilon_j^2}).$$

Using $\prod_i (1 - a_i) \geq 1 - \sum_i a_i$ for all $a_i \in [0, 1]$, and considering the last iteration N,

$$\Pr(H_N) \geq 1 - \sum_{q=1}^{N} 2e^{-2m_q\epsilon_q^2}$$

□

A.2 Uniform Accuracy for Swap-and-Grow

Lemma 3.3 Let $\delta > 0$ be given. Consider Algorithm 1, which evaluates a sequence of candidate sets $S_1, S_2, \ldots, S_N$ chosen adaptively during its execution. (Formally, at any q, the choice of S_t depends on $S_1, S_2, \ldots,$ as well as the estimates $\widehat{\theta}(S_i)$, for $i < q$.) Then if each query is evaluated using $m = \frac{8}{\delta^2}((2t + 6)\ln n + ln(2(t + 1)))$ samples, with probability at least $1 - \frac{1}{n^4}$, we have $|\theta(S_q) - \widehat{\theta}(S_q)| < \frac{\delta}{4}$ for *all* $1 \leq q \leq N$.

Proof of Lemma 3.3. At cardinality level ℓ, algorithm 1 the algorithm evaluates all $(s + 1)$-for-s candidates for $s \in \{0, ..., \min\{t, \ell\}\}$. Therefore, $|C_\ell| \leq \sum_{s=0}^{\min\{t,\ell\}} \binom{\ell}{s}\binom{n-\ell}{s+1}$. Using $\binom{\ell}{s} \leq n^s$ and $\binom{n-\ell}{s+1} \leq n^{s+1}$, we get: $|C_\ell| \leq \sum_{s=0}^{t} n^{2s+1} \leq (t + 1)n^{2t+1}$. Summing over all n levels, $\sum_{\ell=0}^{n-1} |C_\ell| \leq (t + 1)n^{2t+2}$. Therefore, total number of iteration is $N \leq (t + 1)n^{2t+2} = M_t$.

Now, from lemma A.1, we know that,

$$\Pr(\forall q \in [N] : |\widehat{\theta}(S_q) - \theta(S_q)| \leq \epsilon_q) \geq 1 - \sum_{q=1}^{N} 2e^{-2m_q\epsilon_q^2},$$

where m_q is the sample used by set S_q and ϵ_q is its tolerance level.

In case of Algorithm 1, we consider $m_q = m$ and $\epsilon_q = \delta/4$. Therefore, the failure probability is at most:

$$\sum_{q=1}^{N} 2e^{-2m(\frac{\delta}{4})^2} = 2Ne^{-2m(\frac{\delta}{4})^2} \leq 2(t + 1)n^{2t+2}e^{-2m(\frac{\delta}{4})^2} \leq \frac{1}{n^4},$$

and therefore,

$$m \geq \frac{8}{\delta^2}((2t + 6)\ln n + \ln 2(t + 1))$$

□

A.3 Uniform Accuracy for Grow-and-Tune

Lemma 4.4. Let $\delta > 0$ be given. Consider the Algorithm 4 under Bernoulli subset-level feedback model adaptively queries $\theta(S)$ values for a sequence of sets $S_1, S_2, \ldots, S_N$. (Formally, at any q, the choice of S_t depends on $S_1, S_2, \ldots,$ as well as the estimates $\widehat{\theta}(S_i)$, for $i < q$.) Then if each query is evaluated using $\frac{8\ell^2}{\delta^2\gamma^2}\ln(2n^5[(n - \ell + 1) + \ell(n - \ell)(1 + \frac{1}{\eta})])$ samples, with probability at least $1 - \frac{1}{n^4}$, we have $|\theta(S_q) - \widehat{\theta}(S_q)| < \frac{\delta\gamma}{4\ell}$ for *all* $1 \leq q \leq N$.

Proof of Lemma 4.4. In the strictly monotone algorithm, we let C_ℓ denote the collection of all candidate subsets evaluated during level ℓ, including the grow candidates and all swap candidates considered during the tuning steps. At a fixed level ℓ, the grow step considers $n - \ell$ candidates. In

each tuning round, the single-swap neighborhood has size $\ell(n-\ell)$, and the algorithm accepts at most $1/\eta$ swaps in level ℓ. Hence,

$$|C_\ell| \le (n-\ell) + \left(\frac{1}{\eta}+1\right)\ell(n-\ell) = (n-\ell)\left(1+\ell+\frac{\ell}{\eta}\right).$$

Now, from lemma A.1, we know that,

$$\Pr(\forall q \in [N] : |\widehat{\theta}(S_q) - \theta(S_q)| \le \epsilon_q) \ge 1 - \sum_{q=1}^{N} 2e^{-2m_q\epsilon_q^2},$$

As we have to ensure different accuracy tolerance ϵ_ℓ based on which level the set S_q belongs to, we partition $[N]$ into $I_0, ..., I_{n-1}$, where $|I_\ell| = |C_\ell|$. Thus, we consider $m_q = m_\ell$ and $\epsilon_q = \frac{\delta\gamma}{4\ell}$ for all $q \in I_\ell$. Now, we partition the sum of the failure probability by level: $\sum_{q=1}^{N} 2e^{-2m_q(\frac{\delta\gamma}{4\ell})^2} = \sum_{\ell=0}^{n-1}\sum_{q\in I_\ell} 2e^{-2m_\ell(\frac{\delta\gamma}{4\ell})^2} = \sum_{\ell=0}^{n-1} 2|C_\ell|\, e^{-2m_\ell(\frac{\delta\gamma}{4\ell})^2}$.

Now, if we bound over failure probability at a level by $1/n^5$, we get, $2|C_\ell|\, e^{-2m_\ell(\frac{\delta\gamma}{4\ell})^2} \le \frac{1}{n^5}$, and therefore

$$m_\ell \ge \frac{8\ell^2}{\delta^2\gamma^2}\ln(2n^5(n-\ell)(1+\ell+\frac{\ell}{\eta})).$$

A final union bound over all n level shows that the total failure probability is at most $n\cdot\frac{1}{n^5} = \frac{1}{n^4}$. □

B Additional GS Preliminaries

B.1 Consequence of Gradient-Smoothness

Lemma 2.3. Suppose f is a monotone decreasing function that satisfies the GS property with parameters (Δ,γ) and let $I = [a, a+\Delta]$ be an interval. Then for any $x < y < z \in I$, we have:

$$f(y) \ge f(z) + \frac{\gamma(z-y)}{(z-x)}(f(x) - f(z)).$$

Proof of Lemma 2.3. The proof is fairly straightforward, using the definition of the GS property. First, note that by the intermediate value theorem, there exists $w \in [x,z]$ such that $|f'(w)| \ge \frac{(f(x)-f(z))}{(z-x)}$. This implies that for *all* $w \in [x,z]$, we have

$$|f'(w)| \ge \gamma\cdot\frac{(f(x)-f(z))}{(z-x)}.$$

Since f is monotone decreasing, the derivative is always negative, implying that

$$f(z) \le f(y) - \frac{\gamma(z-y)}{(z-x)}\cdot(f(x)-f(z)).$$

Rearranging, the lemma follows. □

C Additional Proofs for Weakly Monotone Safety Functions

C.1 Persistence and Offline Approximation

Lemma 3.2. Assume $r_i \ge 0$ for all $i \in \mathcal{T}$. Let $A, B \subseteq \mathcal{T}$, with $A \ne \emptyset$, and suppose $|A| \ge \frac{t+1}{t}|B|$. Then there exists a set $B^+ \in \Gamma_t(B)$ such that

$$R(B^+) \le \max\{R(A), R(B)\}.$$

Consequently, since f is non-increasing,

$$\theta(B^+) \ge \min\{\theta(A), \theta(B)\}.$$

Proof of Lemma 3.2. Let

$$A_0 := A \setminus B, \qquad B_0 := B \setminus A, \qquad C := A \cap B,$$

and write

$$q := |A_0|, \qquad d := |B_0|, \qquad c := |C|.$$

The size assumption gives

$$c + q = |A| \geq \frac{t+1}{t}|B| = \frac{t+1}{t}(c + d).$$

Hence

$$q \geq \frac{t+1}{t}d + \frac{1}{t}c \geq \frac{t+1}{t}d.$$

First suppose $d = 0$. Then $B \subseteq A$. Since

$$|A| \geq \frac{t+1}{t}|B| > |B|$$

whenever $B \neq \emptyset$, and since $A \neq \emptyset$ if $B = \emptyset$, there is an element $v \in A \setminus B$. Taking

$$B^+ := B \cup \{v\}$$

gives $B^+ \in \Gamma_t(B)$. Moreover $B^+ \subseteq A$, so by nonnegativity of the risks,

$$R(B^+) \leq R(A) \leq \max\{R(A), R(B)\}.$$

Now suppose $d \geq 1$. Let

$$s := \min\{t, d\}.$$

We claim that $q \geq s + 1$. If $d < t$, then $s = d$ and

$$q \geq \frac{t+1}{t}d > d,$$

so, since q is an integer, $q \geq d + 1 = s + 1$. If $d \geq t$, then $s = t$ and

$$q \geq \frac{t+1}{t}d \geq t + 1 = s + 1.$$

Thus there are at least $s + 1$ elements in A_0.

Let U be the s largest-risk elements of B_0, and let V be the $s + 1$ smallest-risk elements of A_0. Define

$$B^+ := (B \setminus U) \cup V.$$

Then $B^+ \in \Gamma_t(B)$. By the averaging choice of U and V,

$$R(U) \geq \frac{s}{d}R(B_0), \qquad R(V) \leq \frac{s+1}{q}R(A_0).$$

Since $q \geq \frac{t+1}{t}d$ and $s = \min\{t, d\}$, we have

$$\frac{s+1}{q} \leq \frac{s}{d}.$$

Let

$$\lambda := \frac{s}{d}.$$

Then

$$\begin{aligned} R(B^+) &= R(B) - R(U) + R(V) \\ &\leq R(C) + R(B_0) - \lambda R(B_0) + \lambda R(A_0) \\ &= (1 - \lambda)R(B) + \lambda R(A). \end{aligned}$$

Since $\lambda \in [0, 1]$, this convex combination is at most $\max\{R(A), R(B)\}$. This proves the risk bound.

Finally, because f is non-increasing and

$$R(B^+) \le \max\{R(A), R(B)\},$$

we get

$$\theta(B^+) = f(R(B^+)) \ge f(\max\{R(A), R(B)\}) = \min\{\theta(A), \theta(B)\}.$$

□

Lemma 3.4. Assume that $\mathcal{E}$ from Lemma 3.3 holds. Fix a non-empty reference set $A \subseteq \mathcal{T}$. For every cardinality level ℓ satisfying $|A| \ge \frac{t+1}{t}\ell$, the representative chosen by algorithm 1 satisfies

$$\theta(S_{\ell+1}) \ge \min\{\theta(A), \theta(S_\ell)\} - \frac{\delta}{2}.$$

Furthermore, suppose we define $L_A := \lfloor \alpha|A| \rfloor + 1$. Then representatives $S_0, S_1, \ldots, S_n$ constructed by Algorithm 1 satisfies

$$\theta(S_\ell) \ge \theta(A) - \frac{\delta\ell}{2} \qquad \text{for every } 0 \le \ell \le L_A.$$

Proof of Lemma 3.4. Fix a level ℓ satisfying the cardinality condition and apply Lemma 3.2 with $B = S_\ell$. Since $|S_\ell| = \ell$, there exists a candidate $B^+ \in \Gamma_t(S_\ell) = C_\ell$ such that $\theta(B^+) \ge \min\{\theta(A), \theta(S_\ell)\}$. Since $S_{\ell+1}$ maximizes the empirical safety estimate over C_ℓ, $\widehat{\theta}(S_{\ell+1}) \ge \widehat{\theta}(B^+)$. On $\mathcal{E}$,

$$\theta(S_{\ell+1}) \ge \widehat{\theta}(S_{\ell+1}) - \frac{\delta}{4} \ge \widehat{\theta}(B^+) - \frac{\delta}{4} \ge \theta(B^+) - \frac{\delta}{2} \ge \min\{\theta(A), \theta(S_\ell)\} - \frac{\delta}{2},$$

which proves (1).

For the "furthermore" part of the Lemma, we use induction on ℓ. The case $\ell = 0$ is trivial: we have $S_0 = \emptyset$. Since risks are nonnegative, $R(S_0) = 0 \le R(A)$, and thus $\theta(S_0) \ge \theta(A)$. Next, suppose the claim holds at level ℓ, where $0 \le \ell < L_A$. Since

$$\ell \le L_A - 1 = \lfloor \alpha|A| \rfloor \le \alpha|A|,$$

we have $|A| \ge \frac{t+1}{t}\ell$. Thus using the first part of the Lemma,

$$\theta(S_{\ell+1}) \ge \min\{\theta(A), \theta(S_\ell)\} - \frac{\delta}{2}.$$

By the induction hypothesis, we have $\theta(S_\ell) \ge \theta(A) - \frac{\delta\ell}{2}$. Hence,

$$\theta(S_{\ell+1}) \ge \min\{\theta(A), \theta(S_\ell)\} - \frac{\delta}{2} \ge \theta(A) - \frac{\delta(\ell+1)}{2}.$$

This completes the proof by induction. □

Theorem 3.1. Let $S^\star \subseteq \mathcal{T}$ maximizes $\mathrm{Obj}(S)$, and let $\epsilon > 0$. Run the algorithm 1 with $\delta := \frac{\alpha\epsilon}{2n^2}$. Then with probability at least $1 - \frac{1}{n^4}$, its output $\widehat{S}$ satisfies:

$$\mathrm{Obj}(\widehat{S}) \ge \frac{t}{t+1} \cdot (\mathrm{Obj}(S^\star) - \epsilon).$$

Moreover, the number of noisy queries made by the algorithm is $\mathrm{poly}(n^t)/\epsilon^2$.

Proof of Theorem 3.1. Let $\alpha = \frac{t}{t+1}$ as before, and let $S^\star$ be an optimal set (one that maximizes the objective). Write

$$\ell^\star := |S^\star|, \qquad \theta^\star := \theta(S^\star), \qquad \mathrm{OPT} := \mathrm{Obj}(S^\star) = \ell^\star\theta^\star.$$

If $\ell^\star = 0$ or $\mathrm{OPT} = 0$, the claim is immediate because objectives are nonnegative. Hence assume $\ell^\star \ge 1$. Now suppose we run the algorithm with enough samples such that event $\mathcal{E}$ holds with

probability $1 - n^{-4}$ from Lemma 3.3. Define $k := \lceil \alpha \ell^\star \rceil$. Since $k \le \lfloor \alpha \ell^\star \rfloor + 1$, we apply Lemma 3.4 with $A = S^\star$ to obtain

$$\theta(S_k) \ge \theta^\star - \frac{k\delta}{2}.$$

Moreover, $k \ge \alpha \ell^\star$. Therefore

$$\mathrm{Obj}(S_k) \ge k\left(\theta^\star - \frac{k\delta}{2}\right) \ge \alpha \ell^\star \theta^\star - \frac{\delta}{2}k^2 \ge \alpha \mathrm{OPT} - \frac{\delta n^2}{2}.$$

It remains only to account for the final empirical selection. On $\mathcal{E}$, for every representative S_ℓ, $\left||S_\ell|\widehat{\theta}(S_\ell) - \mathrm{Obj}(S_\ell)\right| \le \frac{n\delta}{4}$. Since $\widehat{S}$ maximizes empirical objective over the constructed chain,

$$\begin{aligned}
\mathrm{Obj}(\widehat{S}) &\ge |\widehat{S}|\widehat{\theta}(\widehat{S}) - \frac{n\delta}{4} \ge k\widehat{\theta}(S_k) - \frac{n\delta}{4} \ge \mathrm{Obj}(S_k) - \frac{n\delta}{2} \\
&\ge \alpha \mathrm{OPT} - \frac{\delta n^2}{2} - \frac{n\delta}{2} \\
&\ge \alpha \mathrm{OPT} - \frac{\delta}{2} n(n+1).
\end{aligned}$$

With our choice $\delta = \alpha\epsilon/(2n^2)$, the error term above can be bounded by $\alpha\epsilon$. Hence

$$\mathrm{Obj}(\widehat{S}) \ge \alpha \mathrm{OPT} - \alpha\epsilon = \frac{t}{t+1}\left(\mathrm{Obj}(S^\star) - \epsilon\right).$$

The good event holds with probability at least $1 - n^{-4}$ by Lemma 3.3, thus completing the proof. □

C.2 Exact-Oracle Lower Bound

Theorem 3.5. For any fixed $t \ge 1$ and all sufficiently large n, any randomized algorithm $\mathcal{A}$ with query complexity $< \frac{1}{10}\binom{n}{t+1}$, there exists a weakly monotone set function θ such that: (i) for the optimal set $S^\star$, we have $\mathrm{Obj}(S^\star) \ge (t+1)$, and (ii) with probability at least $3/4$, the output S of the algorithm satisfies $\mathrm{Obj}(S) \le t$.

Proof of Theorem 3.5. The proof follows the standard paradigm of using Yao's minimax lemma to reduce to the case of deterministic $\mathcal{A}$. Specifically, we give a distribution over monotone set functions θ such that (i) for all θ in the support, we have a set $S^\star$ such that $\mathrm{Obj}(S^\star) = (t+1)$, and (ii) for any *deterministic* algorithm $\mathcal{A}$, with probability at least $3/4$, the output S of the algorithm achieves $\mathrm{Obj}(S) \le t$.

To define a distribution over θ, we first draw the parameters $(r_1, r_2, \ldots, r_n)$ from a distribution, and θ is defined as

$$\theta(S) = \mathbf{1}[\sum_{i \in S} r_i \le t(t+1)].$$

The parameters $(r_1, r_2, \ldots, r_n)$ are sampled as follows: a random subset J of indices is chosen with $|J| = (t+1)$, and we set $r_i = t$ for $i \in J$, and $r_i = (t+1)$ otherwise.

The nice feature of the setting above is the following. If $|S| \le t$, we have $\theta(S) = 1$, and if $|S| \ge (t+2)$, we have $\theta(S) = 0$. When $|S| = (t+1)$, we then look at $x = |S \cap J|$. We have $R(S) = tx + (t+1)(t+1-x) = (t+1)^2 - x$. If $x = (t+1)$, then $\theta(S) = 1$, and if $x \le t$, we have $R(S) \ge t^2 + t + 1$, and thus $\theta(S) = 0$. Thus, when $|S| = (t+1)$, the result for the query $\theta(S)$ is 0 for all S except $S = J$.

Now consider any deterministic algorithm $\mathcal{A}$ that makes at most $\frac{1}{10}\binom{n}{t+1}$ queries. With probability at least $9/10$, $S = J$ will not be one of the queries (even assuming the worst case of the algorithm only querying $(t+1)$-sized sets). Since the algorithm is deterministic, its output in *all the cases* where $S = J$ is not one of the queries has to be the same. Since this fixed output can equal J for at

most one additional choice of J, the probability of outputting J is at most $1/10 + 1/\binom{n}{t+1} \le 1/4$ for sufficiently large n. (And thus the objective is either 0 or $\le t$.) This completes the proof. □

C.3 Singleton Scale and Its Estimation

Lemma 3.7. Let $S^\star$ be an optimal set, and write $\theta^\star := \theta(S^\star)$. If $r_i \ge 0$ for all $i \in \mathcal{T}$, then

$$g \le \mathrm{OPT} = |S^\star|\theta^\star \le ng, \qquad \text{and consequently} \qquad \theta^\star \ge \frac{g}{n}.$$

Proof. The lower bound holds because the best singleton is a feasible set. For the upper bound, because of the nonnegativity of the risks, we have for any $S \subseteq \mathcal{T}$ and $i \in S$, $R(\{i\}) \le R(S)$, which implies that $\theta(\{i\}) \ge \theta(S)$. Hence $\theta(S) \le g$, and therefore $\mathrm{Obj}(S) = |S|\theta(S) \le ng$. Taking the maximum over S proves that $\mathrm{OPT} \le ng$. Moreover, $g \le \mathrm{OPT} = |S^\star|\theta^\star \le n\theta^\star$ implies $\theta^\star \ge \frac{g}{n}$. □

Lemma 3.8. Let $g_0 \in (0,1)$ be a given parameter, and let $g := \max_{i\in\mathcal{T}} \theta(\{i\})$ be the singleton-scale (which is unknown) of the given instance. There exists an algorithm that makes noisy queries as before, and outputs either a real number $\widehat{g}$ or `small`, such that the following hold with probability at least $1 - T^{-2}$:

(1) If the algorithm returns `small`, then $g \le 2g_0$.
(2) If the algorithm returns a value $\widehat{g}$, then $\frac{g}{2} \le \widehat{g} \le 2g$ and also, $g \ge c_0 g_0$, for some absolute constant c_0.

Finally, the number of queries made by the algorithm is at most

$$Q_{\mathrm{sing}} = \widetilde{O}\left(\frac{n}{\max\{g, g_0\}^2}\right).$$

Proof of Lemma 3.8. Consider the following simple procedure (Algorithm 6):

Algorithm 6: SingletonScale(g_0, T)

Let $J := \lfloor \log_2(1/g_0) \rfloor$, so that $2^{-J} \in [g_0, 2g_0]$
for $j = 0, 1, \ldots, J$ **do**
  Set $a_j := 2^{-j}$
  Sample every singleton $\{i\}$,

$$h_j := \left\lceil \frac{32\log(8n(J+1)T^2)}{a_j^2} \right\rceil$$

  times, and let $\widehat{\theta}_j(\{i\})$ be its empirical mean.
  Set $\widehat{g}_j := \max_{i\in\mathcal{T}} \widehat{\theta}_j(\{i\})$
  if $\widehat{g}_j \ge 3a_j/4$ **then**
    return $\widehat{g} := \widehat{g}_j$
return small

At stage j, Hoeffding's inequality gives, for each singleton,

$$\Pr\left(\left|\widehat{\theta}_j(\{i\}) - \theta(\{i\})\right| > a_j/8\right) \le 2\exp\left(-2h_j(a_j/8)^2\right).$$

By the choice of h_j and a union bound over all n singletons and all $J+1$ stages, with probability at least $1 - T^{-2}$, every singleton estimate at every stage is accurate to within $a_j/8$. We condition on this event.

Suppose the algorithm returns at stage j. Then

$$\widehat{g}_j \ge \frac{3a_j}{4}.$$

Accuracy implies

$$g \geq \widehat{g}_j - \frac{a_j}{8} \geq \frac{5a_j}{8},$$

so $a_j \leq \frac{8}{5}g$. Therefore

$$\widehat{g}_j \leq g + \frac{a_j}{8} \leq g + \frac{g}{5} \leq 2g.$$

Also, since $\widehat{g}_j$ is the largest empirical singleton mean,

$$\widehat{g}_j \geq g - \frac{a_j}{8} \geq g - \frac{g}{5} \geq \frac{g}{2},$$

after adjusting constants. Finally, because the algorithm only returns at a scale $a_j \geq 2^{-J} \geq g_0$, the lower bound $g \geq (5/8)a_j$ implies $g \geq c_0 g_0$.

Now suppose the algorithm returns small. Then it did not stop at stage J, where $a_J \in [g_0, 2g_0]$. Hence

$$\widehat{g}_J < \frac{3a_J}{4}.$$

By accuracy,

$$g \leq \widehat{g}_J + \frac{a_J}{8} < \frac{7a_J}{8} \leq 2g_0.$$

The sample complexity is dominated by the last scale reached. If $g \geq g_0$, the procedure stops once $a_j = \Theta(g)$; otherwise it reaches $a_j = \Theta(g_0)$. Therefore

$$Q_{\text{sing}} = \widetilde{O}\left(\frac{n}{\max\{g, g_0\}^2}\right).$$

□

C.4 Calibration and Approximate Regret

Lemma 3.9. Fix $\xi > 0$. Suppose Algorithm 1 samples every candidate set $m_\xi := \left\lceil \frac{\log(4M_t T^2)}{2\xi^2} \right\rceil$ times. Then, with probability at least $1 - T^{-2}/2$, all empirical estimates used by Algorithm 1 are accurate to within ξ. Moreover, the total number of samples used by Algorithm 1 is at most

$$Q_{\text{SG}}(\xi) \leq M_t m_\xi = O\left(\frac{(t+1)n^{2t+2}\log(nT)}{\xi^2}\right).$$

Under the same event, if $\xi \leq \frac{g}{32n^3(t+1)}$, then the output $\widehat{S}$ of Algorithm 1 satisfies

$$\text{Obj}(\widehat{S}) \geq \alpha\text{OPT} + \frac{g}{2n(t+1)}.$$

Proof of Lemma 3.9. For any fixed queried set S, the adaptive concentration bound in Lemma A.1 gives

$$\Pr\left(|\widehat{\theta}(S) - \theta(S)| > \xi\right) \leq 2\exp(-2m_\xi \xi^2).$$

Taking a union bound over at most M_t queried candidates by algorithm 1 gives

$$\Pr(\text{some estimate fails}) \leq 2M_t \exp(-2m_\xi \xi^2) \leq \frac{1}{2T^2}.$$

The sample complexity bound follows immediately from the definition of m_ξ. We condition on this uniform-accuracy event for the reminder of the proof. Now, let $S^\star$ be an optimal set and write

$\ell^\star := |S^\star|$ and $\theta^\star := \theta(S^\star)$. Uniform accuracy to within ξ corresponds to the event of Lemma 3.6 with $\delta = 4\xi$. Applying Lemma 3.6 with $A = S^\star$ gives,

$$\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{OPT} + \frac{\theta^\star}{t+1} - 4\xi n^2.$$

By Lemma 3.7, $\theta^\star \geq \frac{g}{n}$, and therefore,

$$\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{OPT} + \frac{g}{n(t+1)} - 4\xi n^2.$$

If $\xi \leq \frac{g}{32n^3(t+1)}$, then $4\xi n^2 \leq \frac{g}{8n(t+1)}$.
Hence

$$\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{OPT} + \frac{7g}{8n(t+1)} \geq \alpha\mathrm{OPT} + \frac{g}{2n(t+1)}.$$

□

Theorem 3.10. For $\alpha = \frac{t}{t+1}$, the α-regret over horizon T of Algorithm 2 satisfies

$$\mathcal{R}_\alpha(T) = \widetilde{O}((t+1)^{3/2} n^{t+5} \sqrt{T})$$

Equivalently, for fixed t,

$$\mathcal{R}_\alpha(T) = \widetilde{O}_t(n^{t+5}\sqrt{T}).$$

Proof of Theorem 3.10. Let $B_t = C(t+1)^3 n^{2t+9} \log(nT)$ with C large enough to absorb the constants in the sample-complexity bounds. First suppose $g_0 \geq 1$. Then $nT \leq B_t$. Since the regret in any round is at most n,

$$\mathcal{R}_\alpha(T) \leq nT \leq \sqrt{nB_tT}.$$

Thus the desired bound holds. Henceforth assume $g_0 < 1$. Let $\mathcal{G}$ be the event that both the singleton-scale estimator succeeds and, if Algorithm 1 is run (within Alg. 2), all estimates inside Algorithm 1 are accurate to within the prescribed value of ξ. By Lemmas 3.8 and 3.9 we get $\Pr(\mathcal{G}^c) \leq \frac{2}{T^2}$. Therefore, on $\mathcal{G}^c$, the regret is at most nT, so the failure contribution to expected regret is at most $\frac{2}{T^2} \cdot nT = \frac{2n}{T}$. We therefore condition on $\mathcal{G}$.

Case 1: the estimator returns small. By Lemma 3.8, $g \leq 2g_0$ and by Lemma 3.7, $\mathrm{OPT} \leq ng \leq 2ng_0$. Since rewards are nonnegative, over the horizon

$$\mathcal{R}_\alpha(T) \leq \mathrm{OPT}T \leq 2ng_0T = 2nT\sqrt{\frac{B_t}{nT}} = 2\sqrt{nB_tT}.$$

Case 2: the estimator returns $\widehat{g}$. By Lemma 3.8, $\frac{g}{2} \leq \widehat{g} \leq 2g$ and $g \geq c_0 g_0$. Algorithm 2 sets $\xi = \frac{\widehat{g}}{64n^3(t+1)} \leq \frac{g}{32n^3(t+1)}$. Hence, Lemma 3.9 implies

$$\mathrm{Obj}(\widehat{S}) \geq \alpha\mathrm{OPT} + \frac{g}{2n(t+1)}.$$

Therefore every committed round has nonpositive α-regret:

$$\alpha\mathrm{OPT} - \mathrm{Obj}(\widehat{S}) \leq -\frac{g}{2n(t+1)} \leq 0.$$

Hence it suffices to bound the regret incurred during the singleton-estimation and swap-and-grow exploration phases. By Lemma 3.7, during any exploration round, the regret is at most

$$\alpha\mathrm{OPT} \leq \mathrm{OPT} \leq ng.$$

The singleton-estimation phase uses $Q_{\text{sing}} = \widetilde{O}\left(\frac{n}{g^2}\right)$ queries in this case, since $g \geq c_0 g_0$. Its regret contribution is therefore

$$ng\, Q_{\text{sing}} = \widetilde{O}\left(\frac{n^2}{g}\right) \leq \widetilde{O}\left(\frac{n^2}{g_0}\right).$$

Since B_t dominates n^2,

$$\frac{n^2}{g_0} = n^2\sqrt{\frac{nT}{B_t}} \leq \sqrt{nB_tT}.$$

Next, by Lemma 3.9 and the definition of ξ,

$$Q_{\text{SG}} = O\left(\frac{(t+1)n^{2t+2}\log(nT)}{\xi^2}\right) = O\left(\frac{(t+1)^3 n^{2t+8}\log(nT)}{\widehat{g}^2}\right) = O\left(\frac{(t+1)^3 n^{2t+8}\log(nT)}{g^2}\right).$$

Thus the regret contribution of the swap-and-grow exploration phase is at most

$$ng\, Q_{\text{SG}} = O\left(\frac{(t+1)^3 n^{2t+9}\log(nT)}{g}\right) \leq O\left(\frac{B_t}{g}\right) \leq O\left(\frac{B_t}{g_0}\right) = O\left(\sqrt{nB_tT}\right).$$

Combining the two cases and adding the failure contribution gives

$$\mathcal{R}_\alpha(T) \leq \widetilde{O}\left(\sqrt{nB_tT}\right).$$

Finally, substituting $B_t = C(t+1)^3 n^{2t+9}\log(nT)$ yields

$$\mathcal{R}_\alpha(T) = \widetilde{O}\left(\sqrt{n\cdot(t+1)^3 n^{2t+9}\cdot T}\right) = \widetilde{O}\left((t+1)^{3/2} n^{t+5}\sqrt{T}\right).$$

This proves the theorem. □

D Additional Proofs for Strict Offline Optimization

D.1 Exact Oracle Structure

Lemma 4.1. For every $\ell \in \{0, ..., n\}$ and $S_\ell^\star$ defined as above, the set S_ℓ produced by the algorithm 3 satisfies $R(S_\ell) = R(S_\ell^\star)$. Consequently,

$$S_0 \subset S_1 \subset \cdots \subset S_n.$$

is a nested chain of cardinality-level optimal sets.

Proof of Lemma 4.1. The proof is by induction on ℓ. The base case of $\ell = 0$ is trivial. Consider any $\ell \geq 1$. Suppose i_ℓ is the element added to $S_{\ell-1}$ to obtain S_ℓ. Then $R(S_\ell) = R(S_{\ell-1}) + r_{i_\ell}$. As the elements of $\mathcal{T}$ are ordered as $r_{j_1} \leq r_{j_2} \leq \cdots \leq r_{j_n}$, if $r_{i_\ell} = r_{j_\ell}$, then there is nothing to prove. If $r_{i_\ell} < r_{j_\ell}$, then we obtain a contradiction to the definition of $S_\ell^\star$ (i.e., we found a ℓ sized set of strictly largest θ value). Thus, suppose $r_{i_\ell} > r_{j_\ell}$. Let j_l be the least element of $\{j_1, j_2, \ldots, j_\ell\}$ that is *not in* $S_{\ell-1}$. Since $S_{\ell-1}$ has size only $(\ell-1)$, j_l is well defined. By definition, $r_{j_l} \leq r_{j_\ell}$ and we have assumed that $r_{i_\ell} > r_{j_\ell}$. Thus, adding j_l instead of i_ℓ would yield a strictly smaller value of $R(S_\ell)$, which is a contradiction to the definition of i_ℓ (step 2 of the algorithm above), since f is strictly monotone. Thus S_ℓ is optimal at cardinality level ℓ. Since Algorithm 3 constructs S_ℓ by adding one item to $S_{\ell-1}$, the resulting level-optimal sets form a nested chain. □

D.2 One-Swap Risk Reduction

Lemma 4.3. Suppose A and B are two subsets of $\mathcal{T}$ and $|A| = |B| = \ell$. Suppose $R(A) < R(B)$. Then there exist $a \in A$ and $b \in B$ such that after swapping out b for a, i.e., taking $B' = B \setminus \{b\} \cup \{a\}$, we have

$$R(B') \le R(B) - \frac{(R(B) - R(A))}{\ell}.$$

Proof of Lemma 4.3. Let $|A| = |B| = \ell$ and $R(A) < R(B)$. Write the disjoint differences

$$X \coloneqq A \setminus B, \qquad Y \coloneqq B \setminus A,$$

where X contains members exclusive to A and Y contains members exclusive to B. We assume the (common) overlap size $d \coloneqq |A \cap B|$. Then $\ell - d = q \coloneqq |X| = |Y| \le \ell$ and gap in the accumulated risk,

$$R(B) - R(A) = \sum_{y \in Y} r_y - \sum_{x \in X} r_x.$$

Let $x^\star \in \arg\min_{x \in X} r_x$ and $y^\star \in \arg\max_{y \in Y} r_y$. then,

$$q \cdot r_{x^\star} \le \sum_{x \in X} r_x \quad \text{and} \quad q \cdot r_{y^\star} \ge \sum_{y \in Y} r_y.$$

Therefore,

$$r_{y^\star} - r_{x^\star} \ge \frac{R(B) - R(A)}{q} \ge \frac{R(B) - R(A)}{\ell}$$

Consider the swap neighbor $B' \coloneqq (B \setminus \{y^\star\}) \cup \{x^\star\}$. Then $B' \in \mathcal{N}(B)$ and

$$R(B') \;=\; R(B) - r_{y^\star} + r_{x^\star} \;\le\; R(B) - \frac{R(B) - R(A)}{\ell}.$$

Since A is fixed, this is equivalent to $(R(B') - R(A)) \le (1 - \frac{1}{\ell})(R(B) - R(A))$. □

D.3 The Per-Level Invariant

Lemma 4.5. Let S_ℓ be the final representative at level ℓ in Algorithm 4. For all $0 \le \ell \le |S^\star|$, we have

$$\theta(S_\ell) \ge \theta(S_\ell^\star) - \ell\delta.$$

Proof of Lemma 4.5. The lemma will proceed by induction. For $\ell = 0$, there is nothing to prove. Consider any $\ell \le |S^\star|$ and suppose that the conclusion of the Lemma holds for $S_{\ell-1}$.

First, observe that if $\theta(S_\ell) \ge \theta(S_{\ell-1}) - \delta$, then we are already done: we have $\theta(S_\ell) \ge \theta(S^\star_{\ell-1}) - \ell\delta \ge \theta(S^\star_\ell) - \ell\delta$. Thus, we may assume that $\theta(S_\ell) < \theta(S_{\ell-1}) - \delta$.

Next, suppose a is an element in $S^\star \setminus S_{\ell-1}$ that has the smallest value of r_a. Because of $\ell \le |S^\star|$, such an a must exist. For this a, we claim that

$$R(S_\ell) \le R(S_{\ell-1}) + 2r_a. \tag{13}$$

This claim will be important in order to be able to use the gradient smoothness property of f.

Proof of Claim. Consider the set $S'_\ell = S_{\ell-1} \cup \{a\}$. This is one of the candidates considered by the algorithm (indeed, in the first "grow" step of level ℓ). Since the algorithm finishes level ℓ with the set S_ℓ, by definition, we have

$$\widehat{\theta}(S_\ell) \ge \widehat{\theta}(S'_\ell). \tag{14}$$

We also know from before that $\theta(S_\ell) < \theta(S_{\ell-1}) - \delta$. We now consider two cases.

Case 1. $\theta(S'_\ell) \ge \theta(S_{\ell-1}) - \frac{\delta}{2}$.

In this case, $\theta(S'_\ell) > \theta(S_\ell) + \frac{\delta}{2}$, by putting together the observations above. However, this contradicts (14), since we have assumed that all the estimates are accurate up to error $\frac{\delta\gamma}{4\ell} < \frac{\delta}{4}$.

Case 2. $\theta(S'_\ell) < \theta(S_{\ell-1}) - \frac{\delta}{2}$.

Now, by the intermediate value theorem, there exists an $x \in [R(S_{\ell-1}), R(S_{\ell-1}) + r_a]$ such that $|f'(x)| \geq \frac{\delta}{2r_a}$, and thus by the gradient smoothness of f in intervals of size $2r_a \leq 2\bar{r}_a = \Delta^\star$, we have that

$$f(R(S_{\ell-1}) + 2r_a) \leq f(R(S_{\ell-1}) + r_a) - \frac{\delta\gamma}{2} = f(R(S'_\ell)) - \frac{\delta\gamma}{2}.$$

Now, if $R(S_\ell) > R(S_{\ell-1}) + 2r_a$, we have that $f(R(S_\ell)) < f(R(S'_\ell)) - \frac{\delta\gamma}{2}$, and once again, this gives us a contradiction to (14), assuming the accuracy of estimates up to error $\frac{\delta\gamma}{4\ell}$. □

This completes the proof of the Claim, and thus we have (13). We next record the approximate local optimality obtained when the Tune loop terminates. For every $S' \in \mathcal{N}(S_\ell)$, termination implies $\widehat{\theta}(S') \leq \widehat{\theta}(S_\ell) + \eta$. Using $\mathcal{E}$,

$$\theta(S') \leq \theta(S_\ell) + \eta + \frac{\delta\gamma}{2\ell} \leq \theta(S_\ell) + \frac{5\delta\gamma}{8\ell}, \tag{15}$$

where the last inequality follows from $\eta = \frac{\delta\gamma}{8n} \leq \frac{\delta\gamma}{8\ell}$.

We are now ready to complete the proof of the Lemma 4.5. For the sake of contradiction, suppose that $\theta(S^\star_\ell) > \theta(S_\ell) + \ell\delta$. We consider two cases.

Case 1. $R(S_{\ell-1}) \leq R(S^\star_\ell) < R(S_\ell)$.

In this case, the interval $[R(S^\star_\ell), R(S_\ell)]$ has length at most $2r_a$ by the claim above, and so by applying Lemma 4.3, there exists a swap that can be applied to S_ℓ, resulting in S'_ℓ, such that:

$$R(S'_\ell) \leq R(S_\ell) - \frac{(R(S_\ell) - R(S^\star_\ell))}{\ell} \implies \frac{R(S_\ell) - R(S'_\ell)}{R(S_\ell) - R(S^\star_\ell)} \geq \frac{1}{\ell}.$$

Now, we use Lemma 2.3 that follows from gradient smoothness on three sets $S^\star_\ell, S'_\ell, S_\ell$ as we already proved $R(S^\star_\ell) \leq R(S'_\ell) \leq R(S_\ell)$, thus we get,

$$f(R(S'_\ell)) \geq f(R(S_\ell)) + \frac{\ell\delta\gamma}{\ell} = f(R(S_\ell)) + \delta\gamma.$$

This now gives a contradiction to the fact that $\theta(S'_\ell) \leq \theta(S_\ell) + \frac{5\delta\gamma}{8\ell}$, which is eq. (15), and the fact that the estimates are accurate up to an error of $\delta\gamma/4\ell$.

Case 2. $R(S^\star_\ell) < R(S_{\ell-1})$.

In this case, we again have by Lemma 4.3 to conclude that there exists a swap applied to S_ℓ, resulting in S'_ℓ, such that

$$R(S'_\ell) \leq R(S_\ell) - \frac{(R(S_\ell) - R(S_{\ell-1}))}{\ell}.$$

Once again, we can apply Lemma 2.3 to conclude that

$$f(R(S'_\ell)) \geq f(R(S_\ell)) + \frac{\delta\gamma}{\ell}.$$

This gives a contradiction to the fact that $\theta(S'_\ell) \leq \theta(S_\ell) + \frac{5\delta\gamma}{8\ell}$, which is eq. (15), and the fact that estimates are accurate up to an error of $\delta\gamma/4\ell$. □

D.4 Offline Approximation Guarantee

Theorem 4.2. Let $S^\star \subseteq \mathcal{T}$ be the optimal set and let $\epsilon > 0$. Suppose f satisfies the GS property with paramters $(\Delta^\star, \gamma)$, where $\Delta^\star := 2\max_{a\in S^\star} r_a$. Run algorithm 4 with $\delta := \frac{\epsilon}{8n^2}$ and $\eta = \frac{\delta\gamma}{8n}$. Then with probability $\geq 1 - \frac{1}{n^4}$, the output $\widehat{S}$ of Algorithm 4 satisfies the condition:

$$\mathrm{Obj}(\widehat{S}) \geq (\mathrm{Obj}(S^\star) - \epsilon/4).$$

Proof of Theorem 4.2. Let $S^\star$ be a globally optimal set, and write $\ell^\star := |S^\star|$ and $\mathrm{OPT} := \ell^\star\theta(S^\star)$. We condition on the uniform concentration event from Lemma 4.4. This event holds with probability at least $1-1/n^4$ and on this event every estimate used at level ℓ satisfies $|\widehat{\theta}(S)-\theta(S)| \leq \frac{\delta\gamma}{4\ell}$. By Lemma 4.5, the representative constructed at level $\ell^\star$ satisfies $\theta(S_{\ell^\star}) \geq \theta(S^\star_{\ell^\star}) - \ell^\star\delta$. Since $S^\star_{\ell^\star} = S^\star$, this becomes $\theta(S_{\ell^\star}) \geq \theta(S^\star) - \ell^\star\delta$. Multiplying by $\ell^\star$, we obtain

$$\mathrm{Obj}(S_{\ell^\star}) = \ell^\star\theta(S_{\ell^\star}) \geq \ell^\star\theta(S^\star) - \ell^{\star 2}\delta = \mathrm{OPT} - \ell^{\star 2}\delta.$$

Thus, the representative at the optimal cardinality already has objective value close to OPT. It remains to account for the final empirical selection step. Let $\widehat{\ell} \in \arg\max_{\ell\in[n]} \ell\,\widehat{\theta}(S_\ell)$, and let the output of Algorithm 4 be $\widehat{S} := S_{\widehat{\ell}}$. For every level ℓ, the uniform concentration event ensures $\left|\ell\,\widehat{\theta}(S_\ell) - \ell\,\theta(S_\ell)\right| \leq \frac{\delta\gamma}{4}$. Therefore,

$$\mathrm{Obj}(\widehat{S}) = \widehat{\ell}\,\theta(S_{\widehat{\ell}}) \geq \widehat{\ell}\,\widehat{\theta}(S_{\widehat{\ell}}) - \frac{\delta\gamma}{4}.$$

Since $\widehat{\ell}$ maximizes the empirical objective, $\widehat{\ell}\,\widehat{\theta}(S_{\widehat{\ell}}) \geq \ell^\star\,\widehat{\theta}(S_{\ell^\star})$. Hence, $\mathrm{Obj}(\widehat{S}) \geq \ell^\star\,\widehat{\theta}(S_{\ell^\star}) - \frac{\delta\gamma}{4}$. Applying the concentration bound once more to $S_{\ell^\star}$, we obtain $\ell^\star\,\widehat{\theta}(S_{\ell^\star}) \geq \ell^\star\,\theta(S_{\ell^\star}) - \frac{\delta\gamma}{4}$. Combining the previous two inequalities gives

$$\mathrm{Obj}(\widehat{S}) \geq \mathrm{Obj}(S_{\ell^\star}) - \frac{\delta\gamma}{2} \geq \mathrm{OPT} - \ell^{\star 2}\delta - \frac{\delta\gamma}{2}.$$

We now set $\delta := \frac{\epsilon}{8n^2}$. Since $\ell^\star \leq n$ and $\gamma \leq 1$, $\ell^{\star 2}\delta \leq n^2 \cdot \frac{\epsilon}{8n^2} = \frac{\epsilon}{8}$, and $\frac{\delta\gamma}{2} \leq \frac{\epsilon}{16n^2} \leq \frac{\epsilon}{16}$. Therefore,

$$\mathrm{Obj}(\widehat{S}) \geq \mathrm{OPT} - \frac{\epsilon}{4}.$$

□

E Additional Proofs and Subroutines for Optimistic Grow-and-Tune

E.1 Warm Start Inside the Optimal GS Window

Lemma 4.6 Condition on $\mathcal{E}_{\mathrm{ws}}$. The warm-start representative of the globally optimal cardinality-level, $S^0_{\ell^\star}$ is $\ell^\star$-window good, that is, $R(S^\star_{\ell^\star}) \leq R(S^0_{\ell^\star}) \leq R(S^\star_{\ell^\star}) + 2\bar{r}_{\ell^\star}$, where, $\bar{r}_{\ell^\star} := \max_{a\in S^\star_{\ell^\star}} r_a$. Equivalently, since $S^\star = S^\star_{\ell^\star}$ and $\Delta^\star = 2\max_{a\in S^\star} r_a = 2\bar{r}_{\ell^\star}$, we have

$$R(S^\star) \leq R(S^0_{\ell^\star}) \leq R(S^\star) + \Delta^\star.$$

Proof of Lemma 4.6. For each level $\ell < n$, we define its outside-window gap

$$\omega_\ell := \theta(S^\star_\ell) - \max\left\{\theta(S) : |S| = \ell,\ R(S) > R(S^\star_\ell) + 2\bar{r}_\ell\right\}, \tag{16}$$

where $\bar{r}_\ell := \max_{a\in S^\star_\ell} r_a$, and we use the convention that the maximum over the empty set is $-\infty$. Intuitively, ω_ℓ measures the smallest loss in success probability incurred by choosing a size-ℓ set outside the GS window. Thus it connects ℓ-window goodness which is a property in $R(.)$ to the success probability which is $\theta(.) = f(R(.))$. Therefore, if S^0_ℓ is not ℓ-window-good, then $\theta(S^\star_\ell) - \theta(S^0_\ell) \geq \omega_\ell$.

We first prove the following claim regarding the consequence of a small outside-window gap:
Claim: For every $\ell < \ell^\star$ and $0 < \gamma \le 1$,

$$\theta^\star_\ell - \theta^\star_{\ell+1} \le \left(1 + \frac{1}{\gamma}\right)\omega_\ell, \qquad \text{where } \theta^\star_\ell := \theta(S^\star_\ell) \tag{17}$$

Proof of Claim (17). To prove (17), order the items by nondecreasing risk, $r_{j_1} \le r_{j_2} \le \cdots \le r_{j_n}$. By the nested level-optimal structure,

$$S^\star_\ell = \{j_1, \ldots, j_\ell\}, \qquad S^\star_{\ell+1} = \{j_1, \ldots, j_\ell, j_{\ell+1}\}.$$

Let

$$x := R(S^\star_\ell), \qquad a := \bar{r}_\ell = r_{j_\ell}, \qquad b := r_{j_{\ell+1}}.$$

Then $R(S^\star_{\ell+1}) = x + b$. Let y be the smallest accumulated risk among size-ℓ sets outside the GS window:

$$y := \min\{R(S) : |S| = \ell,\ R(S) > x + 2a\}.$$

If no such set exists, then $\omega_\ell = +\infty$, and (17) is immediate. Otherwise, by monotonicity of f, $\omega_\ell = f(x) - f(y)$.

First suppose that $y \ge x + b$. Since f is decreasing, $f(x) - f(x+b) \le f(x) - f(y) = \omega_\ell$, and hence (17) follows immediately. Therefore, suppose that $y < x + b$, and define $d := x + b - y > 0$.

We claim that $d \le a$. If $b \le 3a$, then $y > x + 2a$, and therefore $d = x + b - y < b - 2a \le a$. Otherwise, if $b > 3a$, consider the size-ℓ set

$$S^\star_{\ell+1} \setminus \{j_\ell\}.$$

Its risk is $x + b - a > x + 2a$, so it lies outside the window. Moreover, every size-ℓ set different from $S^\star_\ell$ has accumulated risk at least $x + b - a$: any such set must replace at least one item of $S^\star_\ell$ by an item outside $S^\star_\ell$, whose risk is at least b, while the largest risk available for removal from $S^\star_\ell$ is a. Hence $y = x + b - a$, and therefore $d = a$. Thus $d \le a$ in all cases.

Now consider the interval

$$I := [y - d, x + b].$$

Its length is $2d \le 2a = 2\bar{r}_\ell$, so the GS property applies on I. The two intervals $[y-d, y]$ and $[y, x+b]$ have the same length d. Hence Gradient-Smoothness gives

$$f(y) - f(x+b) \le \frac{1}{\gamma}\big(f(y-d) - f(y)\big).$$

Also, $y - d \ge x$, and f is decreasing, so $f(y-d) - f(y) \le f(x) - f(y) = \omega_\ell$. Therefore, $f(y) - f(x+b) \le \frac{\omega_\ell}{\gamma}$. Combining this with $f(x) - f(y) = \omega_\ell$, we obtain

$$\theta^\star_\ell - \theta^\star_{\ell+1} = f(x) - f(x+b) = f(x) - f(y) + f(y) - f(x+b) \le \left(1 + \frac{1}{\gamma}\right)\omega_\ell,$$

which proves the claim of (17). □

Now, this claim becomes the heart of our remaining of the proof. In fact, if $\omega_\ell < \frac{\gamma}{1+\gamma} \cdot \frac{\theta^\star_\ell}{\ell+1}$, then $\theta^\star_\ell - \theta^\star_{\ell+1} < \frac{\theta^\star_\ell}{\ell+1}$, and hence $\theta^\star_{\ell+1} > \frac{\ell}{\ell+1}\theta^\star_\ell$. Multiplying by $\ell + 1$ gives $V^\star_{\ell+1} > V^\star_\ell$. In particular, if $V^\star_\ell \ge \frac{g}{2}$, then

$$\theta^\star_\ell = \frac{V^\star_\ell}{\ell} \ge \frac{g}{2\ell}.$$

Since $0 < \gamma \le 1$, $\frac{\gamma}{1+\gamma} \ge \frac{\gamma}{2}$, and therefore $\frac{\gamma}{1+\gamma} \cdot \frac{\theta_\ell^\star}{\ell+1} \ge \frac{\gamma g}{4\ell(\ell+1)} \ge \frac{\gamma g}{4n^2}$. Thus,

$$\omega_\ell < \frac{\gamma g}{4n^2} \quad \Longrightarrow \quad V_{\ell+1}^\star > V_\ell^\star, \tag{18}$$

whenever $V_\ell^\star \ge g/2$. Intuitively, if ω_ℓ is small, then level ℓ is dominated by level $\ell+1$ and therefore level ℓ cannot be globally optimal.

We now apply this to the warm-start representatives. If $\ell^\star = n$, then the only size-n set is $\mathcal{T}$, so $S_n^0 = S_n^\star = \mathcal{T}$, and the result is immediate. Suppose therefore that $\ell^\star < n$, and assume for contradiction that $S_{\ell^\star}^0$ is not $\ell^\star$-window-good. By the definition of (16),

$$\omega_{\ell^\star} \le \theta(S_{\ell^\star}^\star) - \theta(S_{\ell^\star}^0).$$

On the other hand, the warm-start guarantee in (2) gives $\theta(S_{\ell^\star}^\star) - \theta(S_{\ell^\star}^0) \le \ell^\star\, \delta_{\text{init}}$. Hence

$$\omega_{\ell^\star} \le \ell^\star\, \delta_{\text{init}} \le n\delta_{\text{init}} = \frac{\gamma g}{128n^2} < \frac{\gamma g}{4n^2}.$$

Moreover, since $\ell^\star$ is globally optimal, $V_{\ell^\star}^\star = \mathrm{OPT}$. The best singleton has objective g, so $\mathrm{OPT} \ge g$, and therefore $V_{\ell^\star}^\star \ge g > \frac{g}{2}$. Applying (18) with $\ell = \ell^\star$ gives

$$V_{\ell^\star+1}^\star > V_{\ell^\star}^\star,$$

contradicting the definition of $\ell^\star$. Thus $S_{\ell^\star}^0$ must be $\ell^\star$-window-good:

$$R(S_{\ell^\star}^\star) \le R(S_{\ell^\star}^0) \le R(S_{\ell^\star}^\star) + 2\bar{r}_{\ell^\star}.$$

Finally, because $S^\star = S_{\ell^\star}^\star$ and $\Delta^\star = 2\bar{r}_{\ell^\star}$, this is equivalently

$$R(S^\star) \le R(S_{\ell^\star}^0) \le R(S^\star) + \Delta^\star.$$

□

Lemma 4.7. Fix a level ℓ and a size-ℓ set S, where S is ℓ-window good. Under the active Gradient-Smoothness condition on the interval between $S_\ell^\star$ and S,

$$G_\ell(S) \le \frac{\ell^2}{\gamma} \max_{S' \in \mathcal{N}(S)} [\theta(S') - \theta(S)]_+,$$

where $\mathcal{N}(S) := \{(S \setminus \{i\}) \cup \{j\} : i \in S,\ j \notin S\}$ is the single-swap tune neighborhood.

Proof of Lemma 4.7. Let

$$x := R(S_\ell^\star), \qquad z := R(S).$$

Since $S_\ell^\star$ maximizes θ among all size-ℓ sets and f is strictly decreasing, we have $x \le z$. If $x = z$, then $G_\ell(S) = 0$, and the claim follows immediately.

Now suppose $x < z$. Applying Lemma 4.3 with $A = S_\ell^\star$ and $B = S$, there exists a one-swap neighbor $S' \in \mathcal{N}(S)$ such that

$$R(S') \le R(S) - \frac{R(S) - R(S_\ell^\star)}{\ell} = z - \frac{z-x}{\ell}.$$

Write $y := R(S')$. Since S' also has cardinality ℓ, the optimality of $S_\ell^\star$ implies $x \le y$. Combining this with the preceding inequality gives

$$x \le y < z, \qquad \frac{z-y}{z-x} \ge \frac{1}{\ell}.$$

By the assumed Gradient-Smoothness condition on $[x, z]$, we can apply Lemma 2.3 to obtain

$$\begin{aligned}\theta(S') - \theta(S) &= f(y) - f(z) \\ &\geq \gamma \frac{z-y}{z-x}\big(f(x) - f(z)\big) \\ &\geq \frac{\gamma}{\ell}\big(f(x) - f(z)\big) \\ &= \frac{\gamma}{\ell}\big(\theta(S_\ell^\star) - \theta(S)\big).\end{aligned}$$

Recall that

$$G_\ell(S) = \ell\big(\theta(S_\ell^\star) - \theta(S)\big).$$

Therefore,

$$\theta(S') - \theta(S) \geq \frac{\gamma}{\ell^2} G_\ell(S).$$

Since $G_\ell(S) \geq 0$, the improvement is nonnegative. Taking the maximum over all one-swap neighbors yields

$$G_\ell(S) \leq \frac{\ell^2}{\gamma} \max_{S' \in \mathcal{N}(S)} [\theta(S') - \theta(S)]_+,$$

which completes the proof. □

E.2 Online Bernstein Confidence Bounds

Lemma 4.8. With probability at least $1 - T^{-2}$, every set A queried by the algorithm and every time $t \leq T$ at which its estimate is used satisfy

$$\mathrm{LCB}_t(A) \leq \theta(A) \leq \mathrm{UCB}_t(A),$$

and

$$\mathrm{UCB}_t(A) - \mathrm{LCB}_t(A) \leq 2b_t(A).$$

We denote this online confidence event by $\mathcal{E}_{\mathrm{on}}$. For the remainder of the analysis, let $\mathcal{E} := \mathcal{E}_{\mathrm{ws}} \cap \mathcal{E}_{\mathrm{on}}$. .

Proof of Lemma 4.8. For a Bernoulli random variable with mean at most g, Bernstein's inequality implies that, for a sufficiently large constant c,

$$\left|\widehat{\theta}_s(A) - \theta(A)\right| \leq c\left(\sqrt{\frac{gL_T}{s}} + \frac{L_T}{s}\right)$$

simultaneously for all sample counts $s \leq T$, all queried sets, and all times at which the estimates are used, except with probability at most T^{-2}. The clipping at g preserves validity because $\theta(A) \leq g$ for every nonempty set A. The width bound follows directly from the definitions of UCB_t and LCB_t. □

E.3 Helper Subroutines

Algorithm 7: Helper Subroutines for Algorithm 5

Function `ComputeIndex`(ℓ)**:**
- **if** frozen$_\ell$ = true **then**
  - $A_\ell^+ \leftarrow S_\ell, \quad I_\ell^+ \leftarrow 0, \quad D_\ell^- \leftarrow 0, \quad U_\ell \leftarrow \ell\,\mathrm{UCB}(S_\ell) + \varepsilon_{\min}$;
- **else**
  - Compute $\mathcal{N}(S_\ell)$, A_ℓ^+, I_ℓ^+, and D_ℓ^- as in Eq. (5), (6), (7) and (9) respectively.;
  - $U_\ell \leftarrow \ell\,\mathrm{UCB}(S_\ell) + \dfrac{\ell^2}{\gamma} I_\ell^+$;
- **return** U_ℓ;

Function `ClosePhase`(ℓ)**:**
- **if** frozen$_\ell$ = true **then**
  - **return**;
- **while** frozen$_\ell$ = false **do**
  - **if** $\varepsilon_\ell < \varepsilon_{\min}$ **then**
    - frozen$_\ell \leftarrow$ true;
    - **return**;
  - $\varepsilon_\ell^+ \leftarrow \rho_\ell \varepsilon_\ell$;
  - `ComputeIndex`(ℓ);
  - **if** $D_\ell^- \geq \dfrac{\gamma}{8\ell^2}\varepsilon_\ell$ **then**
    - $A_\ell^- \leftarrow \arg\max_{A \in \mathcal{N}(S_\ell)} \mathrm{LCB}(A)$
    - $S_\ell \leftarrow A_\ell^-, \quad \varepsilon_\ell \leftarrow \varepsilon_\ell^+;$ **continue**;
  - **if** $I_\ell^+ \leq \dfrac{\gamma}{\ell^2}\varepsilon_\ell^+$ **then**
    - $\varepsilon_\ell \leftarrow \varepsilon_\ell^+;$ **continue**;
  - **return**;

E.4 Optimal-Level Invariant and Optimism

Lemma 4.9. Condition on $\mathcal{E}$. At every time t, the representative at the optimal level $S_{\ell^\star}$ is $\ell^\star$-window-good and satisfies

$$G_{\ell^\star}(S_{\ell^\star}) \leq \varepsilon_{\ell^\star, k_{\ell^\star}}.$$

Moreover, at every online round,

$$U_{\ell^\star}(t) \geq \mathrm{OPT}.$$

Proof of Lemma 4.9. At initialization, Lemma 4.6 ensures $S_{\ell^\star}(0)$ is $\ell^\star$-window-good. Moreover, by (3),

$$G_{\ell^\star}(S_{\ell^\star}(0)) \leq \mathrm{OPT} \leq \ell^\star g = \varepsilon_{\ell^\star, 0}.$$

We show that both properties are preserved at block advances. Suppose the current scale is $\varepsilon = \varepsilon_{\ell^\star, k_{\ell^\star}}$ and the next scale is $\varepsilon^+ = \rho_{\ell^\star}\varepsilon = \left(1 - \frac{\gamma}{8\ell^\star}\right)\varepsilon$.

If the block advances (closes) without a swap, then $I_{\ell^\star}^+(t) \leq \frac{\gamma}{(\ell^\star)^2}\varepsilon^+$. Since $S_\ell^\star$ is $\ell^\star$-window good, by Lemma 4.7 and the confidence event,

$$G_{\ell^\star}(S_{\ell^\star}) \leq \frac{(\ell^\star)^2}{\gamma} \max_{A \in \mathcal{N}(S_{\ell^\star})} [\theta(A) - \theta(S_{\ell^\star})]_+ \leq \frac{(\ell^\star)^2}{\gamma} I_{\ell^\star}^+(t) \leq \varepsilon^+.$$

The representative is unchanged, so it remains $\ell^\star$-window-good.

If instead the block advances (closes) with a swap, then the accepted neighbor $A^-_{\ell^\star}(t)$ satisfies $\mathrm{LCB}_t(A^-_{\ell^\star}(t)) - \mathrm{UCB}_t(S_{\ell^\star}) \geq \frac{\gamma}{8(\ell^\star)^2}\varepsilon$. On $\mathcal{E}$, this implies the true improvement

$$\theta(A^-_{\ell^\star}(t)) - \theta(S_{\ell^\star}) \geq \frac{\gamma}{8(\ell^\star)^2}\varepsilon.$$

Therefore, after setting $S^+_{\ell^\star} = A^-_{\ell^\star}(t)$,

$$G_{\ell^\star}(S^+_{\ell^\star}) = G_{\ell^\star}(S_{\ell^\star}) - \ell^\star\big(\theta(S^+_{\ell^\star}) - \theta(S_{\ell^\star})\big) \leq \varepsilon - \ell^\star \cdot \frac{\gamma}{8(\ell^\star)^2}\varepsilon = \left(1 - \frac{\gamma}{8\ell^\star}\right)\varepsilon = \varepsilon^+.$$

Moreover, on $\mathcal{E}$, the accepted swap satisfies $\theta(S^+_{\ell^\star}) > \theta(S_{\ell^\star})$. Since f is strictly decreasing, this implies $R(S^+_{\ell^\star}) < R(S_{\ell^\star})$. Because $S^\star$ minimized risk among all size-$\ell^\star$ sets and $S^\star_\ell$ is $\ell^\star$-window good,

$$R(S^\star) \leq R(S^+_{\ell^\star}) \leq R(S_{\ell^\star}) \leq R(S^\star) + \Delta^\star.$$

Thus $S^+_{\ell^\star}$ is again $\ell^\star$-window-good. It remains to prove optimism. If level $\ell^\star$ is non-frozen, then using Lemma 4.7 and $\mathcal{E}$ give

$$\mathrm{OPT} \leq \ell^\star\theta(S_{\ell^\star}) + \frac{(\ell^\star)^2}{\gamma}\max_{A\in\mathcal{N}(S_{\ell^\star})}[\theta(A) - \theta(S_{\ell^\star})]_+ \leq \ell^\star\,\mathrm{UCB}_t(S_{\ell^\star}) + \frac{(\ell^\star)^2}{\gamma}I^+_{\ell^\star}(t) = U_{\ell^\star}(t).$$

If level $\ell^\star$ is frozen, then $G_{\ell^\star}(S_{\ell^\star}) \leq \varepsilon_{\min}$. Hence

$$\mathrm{OPT} \leq \ell^\star\theta(S_{\ell^\star}) + \varepsilon_{\min} \leq \ell^\star\,\mathrm{UCB}_t(S_{\ell^\star}) + \varepsilon_{\min} = U_{\ell^\star}(t).$$

This proves the lemma. □

E.5 Regret Within a Block

Lemma 4.10 Fix a non-frozen block k of level ℓ. Let $N_{\ell,k}$ be the number of sampling rounds spend in the block, and let

$$H_\ell := 1 + |N(S_\ell)| \leq 1 + \ell(n-\ell) \leq n^2$$

denotes the size of its local family. Then on $\mathcal{E}$, the accumulated regret in this block is

$$\mathcal{R}_{\ell,k} \leq \widetilde{O}\Big(\frac{\ell^2}{\gamma}\big|\sqrt{gH_\ell N_{\ell,k}} + H_\ell\big|\Big).$$

Proof of Lemma 4.10. Suppose in round t, the algorithm chooses cardinality-level $\ell = \ell_t$. We consider $S := S_\ell$ and $A := A^+_\ell(t)$. We also define $B_t := b_t(S) + b_t(A)$. Since the algorithm chooses ℓ_t to maximize $U_\ell(t)$, Lemma 4.9 gives $U_\ell(t) \geq U_{\ell^\star}(t) \geq \mathrm{OPT}$. Therefore

$$r_t \leq U_\ell(t) - \ell\theta(Q_t).$$

Because all possible block advances have already been performed before the sample is taken, neither the no-swap nor the swap certificate holds at level ℓ. Let

$$\varepsilon := \varepsilon_{\ell,k_\ell}, \qquad \varepsilon^+ := \rho_\ell\varepsilon.$$

The failure of the no-swap certificate gives $I^+_\ell(t) > \frac{\gamma}{\ell^2}\varepsilon^+$, and the failure of the swap certificate gives, in particular for $A = A^+_\ell(t)$, $\mathrm{LCB}_t(A) - \mathrm{UCB}_t(S) < \frac{\gamma}{8\ell^2}\varepsilon$. Now decompose

$$\begin{aligned} I^+_\ell(t) &= \mathrm{UCB}_t(A) - \mathrm{LCB}_t(S) \\ &= \big(\mathrm{LCB}_t(A) - \mathrm{UCB}_t(S)\big) + \big(\mathrm{UCB}_t(A) - \mathrm{LCB}_t(A)\big) + \big(\mathrm{UCB}_t(S) - \mathrm{LCB}_t(S)\big). \end{aligned}$$

Using the confidence-width bound,

$$I^+_\ell(t) \leq \frac{\gamma}{8\ell^2}\varepsilon + 2b_t(A) + 2b_t(S).$$

Since $\rho_\ell = 1 - \frac{\gamma}{8\ell} \geq \frac{7}{8}$, the previous two displays imply $b_t(S) + b_t(A) \geq c_0 \frac{\gamma}{\ell^2}\varepsilon$ for a universal constant $c_0 > 0$. Substituting back,

$$I_\ell^+(t) \leq C_0 B_t, \qquad \text{for a universal constant } C_0.$$

If $Q_t = S$, then

$$r_t \leq U_\ell(t) - \ell\theta(S) = \ell\big(\mathrm{UCB}_t(S) - \theta(S)\big) + \frac{\ell^2}{\gamma} I_\ell^+(t) \leq 2\ell b_t(S) + C_0 \frac{\ell^2}{\gamma} B_t \leq C \frac{\ell^2}{\gamma} B_t.$$

If $Q_t = A$, then

$$r_t \leq U_\ell(t) - \ell\theta(A) \leq \ell\big(\mathrm{UCB}_t(S) - \mathrm{LCB}_t(A)\big) + \frac{\ell^2}{\gamma} I_\ell^+(t).$$

Moreover, since $I_\ell^+(t) > 0$, $\mathrm{LCB}_t(S) \leq \mathrm{UCB}_t(A)$, and therefore

$$\begin{aligned}\mathrm{UCB}_t(S) - \mathrm{LCB}_t(A) &= \big(\mathrm{UCB}_t(S) - \mathrm{LCB}_t(S)\big) + \big(\mathrm{LCB}_t(S) - \mathrm{UCB}_t(A)\big) + \big(\mathrm{UCB}_t(A) - \mathrm{LCB}_t(A)\big) \\ &\leq 2b_t(S) + 2b_t(A).\end{aligned}$$

Thus the same bound follows:

$$r_t \leq C \frac{\ell^2}{\gamma} B_t, \qquad \text{for a universal constant } C.$$

Now, during a fixed block, the representative S_ℓ and its neighborhood $\mathcal{N}(S_\ell)$ are fixed. The sampled sets therefore belong to a fixed local family of size H_ℓ. In every non-frozen sampling round of this block, the algorithm samples the more uncertain set among S_ℓ and $A_\ell^+(t)$. Hence, if Q_t is the sampled set, $b_t(S_\ell) + b_t(A_\ell^+(t)) \leq 2b_t(Q_t)$. Therefore, the upper-bound of the regret in round t becomes, $r_t \leq C\frac{\ell^2}{\gamma} b_t(Q_t)$. For a fixed set sampled m times, the singleton-scale confidence radius satisfies

$$\sum_{s=1}^{m} \left(\sqrt{\frac{gL_T}{s}} + \frac{L_T}{s} \right) \leq \widetilde{O}\big(\sqrt{gm} + 1\big).$$

Summing over the H_ℓ sets in the local family and applying Cauchy–Schwarz gives

$$\sum_{t \text{ in block}} b_t(Q_t) \leq \widetilde{O}\big(\sqrt{gH_\ell N_{\ell,k}} + H_\ell\big).$$

Multiplying by ℓ^2/γ proves the claim. □

E.6 Total Regret

Theorem 4.11. Assume $g = \max_{a \in \mathcal{T}} \theta(\{a\})$ is known. Let $S^\star$ be an optimal set, $\ell^\star = |S^\star|$, and define $\Delta^\star := 2\max_{a \in S^\star} r_a$. Assume f satisfies the GS property with parameter γ on every interval of length at most $\Delta^\star$. Then the regret of the blocked optimistic Grow-and-Tune algorithm satisfies

$$\mathbb{E}[\mathcal{R}(T)] \leq \widetilde{O}\left(\frac{n^4}{\gamma^{3/2}} \sqrt{gT} + \frac{n^6}{\gamma^2} + \frac{n^{16}}{\gamma^6 g^2} \right).$$

In particular, since $g \leq 1$, and the considering instance-dependent additive term $\mathcal{A}_{\mathrm{inst}}(n, \gamma, g) := \widetilde{O}(\frac{n^6}{\gamma^2} + \frac{n^{16}}{\gamma^6 g^2})$, the regret bound becomes,

$$\mathbb{E}[\mathcal{R}(T)] \leq \widetilde{O}\left(\frac{n^4}{\gamma^{3/2}} \sqrt{T} \right).$$

Proof of Theorem 4.11. We decompose the regret into warm-start, non-frozen, and frozen round contributions:

$$\mathcal{R}(T) \le \mathcal{R}_{\text{warm}} + \mathcal{R}_{\text{nf}} + \mathcal{R}_{\text{fr}}.$$

For the warm start, Algorithm 4 uses $Q_{\text{init}} = \widetilde{O}\left(\frac{n^6}{\delta_{\text{init}}^3 \gamma^3}\right) = \widetilde{O}\left(\frac{n^{15}}{\gamma^6 g^3}\right)$ samples. Since $\text{OPT} \le ng$, the warm-start regret is

$$\mathcal{R}_{\text{warm}} \le ng\, Q_{\text{init}} = \widetilde{O}\left(\frac{n^{16}}{\gamma^6 g^2}\right).$$

We next bound non-frozen regret. Since the scale is initialized as $\varepsilon_{\ell,0} = \ell g$, and gradually reduces upto $\varepsilon_{\min} = \frac{g}{T}$, and also every block multiplies the scale by $\rho_\ell = 1 - \frac{\gamma}{8\ell}$, level ℓ has at most

$$K_\ell \le O(\frac{\ell}{\gamma} \log(\ell T)) = \widetilde{O}\left(\frac{\ell}{\gamma}\right)$$

non-frozen blocks. Let N_ℓ be the total number of non-frozen samples allocated to level ℓ. Summing Lemma 4.10 over its blocks and applying Cauchy–Schwarz,

$$\mathcal{R}_\ell^{\text{nf}} \le \widetilde{O}\left(\frac{\ell^2}{\gamma}\sqrt{\frac{g H_\ell \ell N_\ell}{\gamma}} + \frac{\ell^3 H_\ell}{\gamma^2}\right).$$

Since $H_\ell \le n^2$, and $\sum_{\ell=1}^n N_\ell \le T$, another application of Cauchy–Schwarz gives the final regret for the non-frozen round

$$\mathcal{R}_{\text{nf}} \le \widetilde{O}\left(\frac{n^4}{\gamma^{3/2}}\sqrt{gT} + \frac{n^6}{\gamma^2}\right).$$

It remains to bound frozen rounds. If a frozen level ℓ is selected, then $U_\ell(t) = \ell\, \text{UCB}_t(S_\ell) + \varepsilon_{\min}$. Now, from the consequence of Lemma 4.9 it is obvious that $U_\ell(t) \ge U_{\ell^\star}(t) \ge \text{OPT}$. Therefore, during a round t, playing a frozen-level ℓ, upper bound the instantaneous regret as

$$r_t \le U_\ell(t) - \ell\theta(S_\ell) \le 2\ell b_t(S_\ell) + \varepsilon_{\min}.$$

Summing over all frozen plays, applying the Bernstein-radius summation bound and Cauchy–Schwarz, yields

$$\mathcal{R}_{\text{fr}} \le \widetilde{O}\left(n^{3/2}\sqrt{gT} + n^2\right) + T\varepsilon_{\min}.$$

Since $T\varepsilon_{\min} = g \le 1$, the frozen contribution is dominated by $\mathcal{R}_{\text{nf}}$.

Finally, both the warm-start and online confidence events can be chosen to fail with probability $O(T^{-2})$. On their complement, cumulative regret is at most T OPT $\le ngT$, so the failure contribution to expected regret is negligible. Combining the bounds proves the theorem. □

F Related Work (Detailed)

Our model is closely related to the literature on combinatorial bandits, where the learner selects a subset of available items (often viewed as a "super-arm", where the arms correspond to individual items)-and receives feedback that are determined by the selected set [8, 9]. Much of this literature studies semi-bandit feedback setting, where after selecting a subset the learner observes the outcomes of individual items in the selected subset (e.g., [10, 12, 15, 16, 33, 52]). Such feedback allows the learner to estimate contributions from individual items, which is not possible in our model. The *full-bandit* setting is closer to ours, since feedback is available only at the level of the selected subset, and in literature the version is known to be challenging in general [20, 43, 46]. Classical combinatorial bandits study linear aggregate feedback [8], and Rejwan et al. [46] show how individual arm means can be estimated even when only their sum is observed. Our setting

provides even less information: each query returns only a single Bernoulli outcome with mean $\theta(S) = f(R(S))$, where both the aggregate risk $R(S)$ and the function f are unknown.

Recent work also considers nonlinear rewards under various forms of limited feedback. For example, DART [1] studies fixed-size top-K selection with nonlinear joint rewards under assumptions relating individual-arm quality to the joint reward. Wagde and Saha [49] study top-k identification under an unknown monotone aggregation function, while Wang et al. [50] consider maximum-value rewards under value-index feedback. Other related models include combinatorial logistic bandits [38], which assume a parametric reward model, and single-index bandits [30], which study unknown reward-link functions in a contextual setting. Our problem differs in that we must decide *not only which items to select, but also how many*: adding an item increases potential throughput but may reduce the probability that the entire set succeeds. This tradeoff must be learned from noisy subset-level observations without knowing the underlying risks or the function f. A complementary connection is to combinatorial optimization using comparison oracles. Cohen et al. [13] study linear optimization using only exact comparisons between feasible sets. In our setting, however, these comparisons must be inferred from noisy Bernoulli observations. Moreover, when f is only non-increasing (with only monotonicity assumption), different aggregate risks may yield the same safety probability, making such comparisons less informative.

Our online results are also connected to the literature of approximation regret (α-regret), where the learner competes against a fraction of the optimal objective, as in our first setting under monotonicity alone. This benchmark appears in online learning with approximation algorithms [9, 29], including methods that convert offline approximation procedures into online regret guarantees [21, 42–44, 48]. Related guarantees also arise in online admission control, although against a different benchmark [28]. Local search is particularly relevant to our approach. Fujii [24] studies offline approximation guarantees based on local improvements, while Benadè et al. [6] show how offline local search can be extended to online bandits under a multiplicative local-improvement condition. The main difficulty in our setting is that a suboptimal set *need not have an improving neighbor under our objective Obj*(S). We therefore establish the required local-search guarantees directly from the additive-risk structure, using monotonicity only at first and then with additional Gradient-Smoothness assumption.

Since our problem is motivated by admission control, it is also related to the literature on safe and constrained bandits. Conservative bandits seek to maintain cumulative performance relative to a baseline [31, 51], while safe linear bandits require selected actions to satisfy safety constraints [3, 39, 45]. Other works consider resource budgets over time [5, 14, 53] or safety constraints in combinatorial semi-bandits [26]. Our setting differs in how safety is treated: rather than requiring every selected set to satisfy a separate safety constraint, we maximize its expected throughput, $|S|\theta(S)$, where $\theta(S)$ is the unknown probability that the entire set succeeds.

Finally, our problem is related to resource allocation and learning in shared computing and networking systems, especially where decisions must be made under uncertainty. Prior work studies learning and control using bandits in queueing systems [22, 23, 25, 32], multiresource job scheduling [11], and adversarial network scheduling [17, 27, 35]. Other work considers O-RAN resource allocation [4], online allocation with deadlines and reusable resources [34, 36, 37], and SLA-aware monitoring under limited telemetry [47]. We instead study a *more restrictive and constrained feedback model* in which the controller receives only a binary indication of whether the entire admitted configuration succeeded. For example, an interference or overload alarm may indicate a violation, but reveals neither which items contributed to it nor how close the system was to the safety threshold.